

Should Physicians Choose Their Reimbursement Rate?

Menu Design for Physician Payment Contracts*

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Abstract

Experts can leverage asymmetric information to induce demand for their services, complicating the design of payment contracts. In healthcare, physicians are widely believed to induce either excessive or inadequate treatment. One explanation is that heterogeneous physicians plausibly have different socially optimal contracts while payers typically offer a single contract. I study whether instead offering physicians a budget-neutral menu of contracts can improve welfare. I first develop a model of treatment decisions and show which correlations between physicians' altruism, cost of effort, and productivity are sufficient to improve welfare. I then estimate the model using administrative data on Norwegian primary care. For these physicians, the status quo single contract is inefficient. Physicians prefer a menu and respond by spending more time treating patients. The regulator would value the increase in patient health as equivalent to 5 percent of initial expenditure, with the largest gains for older, chronically ill, and rural patients.

Keywords: social preferences, supplier-induced demand, multidimensional screening

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1 Introduction

A central challenge in healthcare is that physicians almost invariably have private information about the appropriate amount of treatment to provide to a patient. Compared to physicians, patients lack medical expertise, and medical records only summarize physician information to third-party payers. As a result, common reimbursement arrangements may not fully align physicians’ incentives. Payers often reimburse physicians for each unit of treatment (“fee-for-service”). This arrangement can incentivize wasteful spending because a physician may only be willing to accept a marginal payment that exceeds the effective marginal cost. An alternative is to pay physicians an upfront flat fee based on a patient’s expected costs, but removing the financial incentive to spend more time with patients may result in inadequate treatment. Physicians may also have private information about themselves, like an idiosyncratic cost of effort. As a result, the socially efficient incentive structure may vary between physicians, but insurance programs typically use a uniform fee schedule.¹

I present the first empirical evidence that replacing a single physician fee schedule with a shared menu of contracts can improve patient health without increasing spending. A menu allows each physician to choose a combination of a fee-for-service rate and a flat fee. I show that a menu can efficiently separate physicians across multiple fee-for-service rates, depending on the joint distribution of physicians’ three unobserved traits. In the context of Norwegian primary care, I use variation in the fee-for-service rate and patient composition to recover one such distribution. I then derive a budget-neutral and voluntary menu of linear contracts that maximizes patient health.

I quantify the expenditure and health impacts of counterfactual reimbursement schemes using a model of physician decision-making. In the model, physicians choose a reimbursement contract and then treatment hours.² For each patient on a fixed list, a physician chooses treatment hours to maximize a weighted sum of private net income and patient health production (e.g., as in Ellis and McGuire, 1986).³ Drawing on novel evidence from plausibly causal reduced-form research designs, I supplement this model with three types of

¹Fixed administrative fee-for-service rate schedules are employed by public insurers in Australia, Canada, China, Denmark, France, Germany, Japan, Norway, Singapore, Sweden, Switzerland, and Taiwan. These schedules generally cover primary care and sometimes also cover specialist and hospital services. In the United States, 44 percent of healthcare spending is paid by public insurance programs according to a fee schedule and private insurers increasingly negotiate physician reimbursement rates as a multiple of Medicare or Medicaid rates (Gottlieb et al., 2023).

²Each contract consists of a base payment per patient-month and a reimbursement rate per unit of treatment, e.g., an hour of patient interactions. In the United States, Medicare reimburses physicians based on the relative time and difficulty associated with furnishing a Medicare physician fee schedule service, measured as “relative value units.”

³In the baseline model, physicians do not exclude some patients to spend more time with others.

physician traits: altruism is the weight on patient health relative to private profit, cost of effort represents the difficulty of spending time with patients, and productivity scales how patient health responds to treatment intensity. Compared to a regulator, physicians have private information about their traits and patients' initial illness severity. The impact of treatment on patient health is lower when patients have relatively high treatment intensity or low illness severity.

This model provides intuition for why a budget-neutral menu of contracts can increase patient health relative to a default uniform contract. Consider a simplified example with two physicians and a two-contract menu, which consists of a low-rate contract (\$45 per hour and \$1 per patient) and a high-rate contract (\$50 per hour and \$0 per patient). For each physician, raising the rate from \$45 to \$50 has a social value and a private value. Offered a menu, the physician with the larger private value (which exceeds \$1) will choose the higher \$50 rate. The menu outperforms a uniform contract when this physician also has the larger social value. The social value is larger for the physician who works fewer hours at \$45 per hour, because a larger health benefit of incremental hours of treatment and smaller incremental public spending on inframarginal hours.⁴

Depending on the correlation structure of physician heterogeneity, private and social values may be aligned so that a menu of contracts can strictly increase welfare. Continuing the example, suppose that the two physicians vary in cost of effort and altruism. First, an older physician might work fewer hours at \$45 due to trouble hearing patients (high cost), which implies a larger social value of switching to \$50. Second, a rural physician might have long-term relationships with patients (high altruism), leading to a larger private value from switching to \$50, because the rural physician especially values incremental health. Combining these two forces, a menu can efficiently allocate an older rural physician to the high fee-for-service rate and a younger urban physician to the low rate. If instead, one physician has relatively high cost and low altruism, e.g., from being older and urban, then without additional heterogeneity, a menu is unlikely to be efficient. High cost and low altruism both lower treatment hours, so in this case, the private value is small while the social value is large. Likewise, a menu is unlikely to be efficient if physicians only vary along one dimension because a large social value corresponds to a small private value. In the full model, the intuition remains that a menu increases social welfare when social and private values are positively correlated, but that condition does not require a positive correlation between cost and altruism. For example, in the empirical application, high-altruism physicians have low

⁴This simplified example assumes that, e.g., patients and physicians do not vary on other dimensions, patient health increases concavely in treatment hours, only one private and one social value exceeds \$1, and positive finite physician traits.

cost and high productivity.

For a menu to outperform a uniform contract, physicians must be differentiated. I find novel reduced-form evidence that Norwegian primary care physicians vary along multiple dimensions. Consistent with heterogeneity in cost of effort, treatment hours vary widely across observably similar patients, and persistent physician heterogeneity explains a large share of this variation. In the model, a physician with a high cost of effort treats all types of patients less than an otherwise identical physician with a low cost of effort (See Figure A.5 for illustrations). Consistent with heterogeneity in productivity, some physicians cause worse health outcomes among quasi-randomly assigned patients. In the model, a physician with high productivity treats patients more similarly to one another and achieves greater health per treatment hour than an otherwise identical low-productivity physician. Consistent with heterogeneity in altruism, treatment hours change heterogeneously across physicians in response to increased reimbursement rates. In the model, a physician with high altruism responds less when the reimbursement rate increases.⁵ I show theoretically that multi-dimensional heterogeneity is necessary but not sufficient for a menu to increase efficiency. To simulate the effects of counterfactual reimbursement schemes, I need to estimate the joint distribution of physician heterogeneity, particularly its correlation structure.

Norway’s institutional setting and data are particularly well-suited for estimating each physician’s cost of effort, altruism, and productivity. I exploit a large and sudden increase in the reimbursement rate to identify altruism. Local regulations rule out several sources of potentially confounding variation. For example, payment rates are otherwise uniform across physicians, and physicians do not choose their patients. Moreover, the restricted administrative data reflect the public healthcare utilization of nearly all Norwegian residents. I observe individual procedures, detailed demographics, medical histories, and adverse outcomes like avoidable hospitalizations and mortality. To estimate parameters of the structural model, I construct a balanced sample of registered patients and maximize the likelihood of observed treatment hours.⁶ With patient records outside of the estimation sample, I can relax and test assumptions that may be necessary in other settings.⁷

I estimate considerable heterogeneity in physicians’ marginal cost, altruism, productivity,

⁵Intuitively, at any reimbursement rate, altruistic physicians sacrifice profit to further improve patient health, so these physicians have less scope to vary treatment hours.

⁶I focus on total treatment hours rather than subsets of care like procedures or diagnostics, which represent a small share of reimbursement and time. In Norway, primary care physicians screen for illness, manage chronic conditions, approve paid sick leave, and refer patients to specialist and non-emergency hospital services.

⁷For example, I test whether physicians’ hours bunch at capacity constraints, patients systematically sort toward physicians with high health production, physicians with reimbursement rate increases are selected on unobserved traits, and in-sample patients and physicians are nationally representative.

and patient mix. Parameter estimates accurately predict treatment hours across physicians and across time for each physician, even for physicians outside of the estimation sample. The heterogeneity across physicians is meaningful because it corresponds to a high social cost of information asymmetry. Under full information, each physician receives a unique contract. These fee-for-service rates are widely dispersed and incentivize greater treatment hours, increasing welfare by \$8.39 per patient-month over the pre-certification status quo, or approximately 70 percent of baseline spending. Welfare includes patients' benefits from incremental treatment hours, as perceived by physicians. The underlying assumption is that all physicians perceive the same health production function, but physicians differ in the altruism weight placed on health. The health production function maps treatment hours and patient illness severity to a socially relevant measure of patient health. Alternative measures of patient benefits have limitations: for example, inadequate primary care may not measurably increase adverse health outcomes until many years later, in part because outcomes like mortality are rare and highly random. Likewise, due to asymmetric information, patient satisfaction may have a weak relationship with objective treatment quality.

With imperfect information about physician heterogeneity, the optimal menu of contracts still meaningfully increases welfare over the status quo. The difference amounts to \$32 million per year when scaled onto the Norwegian population. For comparison, the best uniform contract improves welfare by \$21 million. The menu consists of seven traded contracts that involve a tradeoff between higher fee-for-service rates and lower base payments. At observed treatment intensity, health production increases in treatment, so higher rates imply greater perceived health for all patients. Consistent with the sufficiency condition, the menu increases welfare because patients with high effective severity are often registered to physicians with high cost of effort and low altruism.

Relative to the status quo, the gains from a menu of linear contracts are striking because menus are rarely featured in physician contract design. Several robustness analyses suggest that welfare improvements are not driven by an idiosyncrasy of the empirical approach or setting. For example, counterfactual outcomes are similar when I incorporate more flexible specifications like preferences for leisure or large perturbations to the estimated joint distribution of physician traits. Shifting from a uniform contract to a menu of contracts might therefore improve outcomes in settings other than Norwegian primary care, particularly when high-need clients lack access to high-intensity providers.

In this paper I synthesize a large theoretical literature on physician contracting into an empirical framework for menu design. In both this paper and the stylized settings featured in prior work, the distribution of physician heterogeneity determines which types of contracts are efficient (Jack, 2005; Choné and Ma, 2011; Naegelen and Mougeot, 2011; Allard,

Jelovac and Léger, 2014; Barham and Milliken, 2014; Wu, Chen and Li, 2017; Wu, 2020; Ji, 2021). I characterize the optimal menu of contracts in terms of parameters that can be estimated with panel variation in reimbursement and cross-sectional variation in patient characteristics. I derive the menu for Norwegian primary care physicians to provide the first empirical evidence that any uniform contract is strictly less efficient. I also extend the literature on socially optimal menu design with multi-dimensional consumer heterogeneity from insurance to an understudied selection market (physician labor supply) while incorporating unique dimensions of heterogeneity (Azevedo and Gottlieb, 2017; Marone and Sabety, 2022; Ho and Lee, 2023). Similar to Marone and Sabety (2022), I estimate a joint distribution of agent types and characterize the relative efficiency of a uniform contract. In a parallel exercise, I use the graphical framework from Einav, Finkelstein and Cullen (2010) to provide intuition for how a two-contract menu can increase efficiency for only some distributions of physicians. Outside of healthcare, there are few studies that empirically evaluate menus of contracts in selection markets, where multi-dimensional heterogeneity is first-order (Belle-mare and Shearer, 2013; D’Haultfoeulle and Février, 2020; Taburet, Polo and Vo, 2024). On the theoretical side, Gottlieb and Moreira (2022) compare the efficiency of a menu versus a uniform contract in a complementary setting.

I simultaneously estimate three key correlated dimensions of heterogeneity, which extends the literature that separately documents variation in physicians’ altruism (Hennig-Schmidt, Selten and Wiesen, 2011; Godager and Wiesen, 2013; Douven, Remmerswaal and Zoutenbier, 2017; Galizzi et al., 2015) and practice style (Epstein and Nicholson, 2009; Chan and Chen, 2026; Doyle, Ewer and Wagner, 2010; Gowrisankaran, Joiner and Léger, 2023). Policies that assume physicians vary along only one dimension may have unintended consequences.⁸ Consistent with prior work, I show that physician treatment decisions respond to financial incentives (Brekke et al., 2017; Einav, Finkelstein and Mahoney, 2018; Eliason et al., 2018; Clemens and Gottlieb, 2014; Cabral, Carey and Miller, 2025; Xiang, 2026). I find heterogeneity in this response, and decompose physician heterogeneity into structural physician types and variation in patient treatment need. This decomposition enables welfare analysis in settings where selection into contracts affects both expenditure and healthcare quality.⁹

My framework emphasizes unobserved patient severity and a menu of linear contracts rather than a non-linear uniform contract. In healthcare settings like primary care, mental health, and dentistry, the regulator cannot observe the socially efficient level of effort and

⁸For example, if an insurer believed that physicians only vary in cost of effort, they might end contracts for physicians with low treatment. However, reimbursing these physicians at higher rates might be more cost-effective, especially if they have lower altruism as well.

⁹In a setting with no effects on healthcare quality, Einav et al. (2021) study hospitals’ selection into bundled contracts and subsequent changes in spending.

instead must rely on altruistic physicians to exercise discretion in allocating effort across patients. In such settings, aligning incentives through differentiated contracts can improve welfare relative to targeting a fixed level of effort for each combination of patient and physician. In related work, Gaynor, Mehta and Richards-Shubik (2023) estimate distributions of cost and altruism of dialysis clinics and derive the optimal non-linear uniform contract for an anti-anemia drug. I extend the framework from that paper to include unobserved patient severity and heterogeneity in productivity. Although a nonlinear uniform contract can achieve greater patient health than a menu of linear contracts, the nonlinear contract is not cost-effective unless it makes most physicians worse off.

In Section 2 I present the theoretical model and show which three correlations between physician traits are sufficient for a menu to improve welfare. In Section 3 I describe the empirical setting and present novel reduced-form evidence consistent with multi-dimensional physician heterogeneity. In Section 4 I discuss the parameterization and identification to recover the estimates, which are summarized in Section 5. In Section 6 I demonstrate the efficiency of a counterfactual menu of contracts, evaluate robustness, and discuss extensions.

2 Theoretical Framework

In this section, I develop a model of physician decision-making to quantify expenditure and health outcomes under counterfactual menus. In the model, physicians have private information about their multi-dimensional types and patients' illness severity. A regulator designs a menu from which each physician chooses a contract. Next, each patient's illness severity is drawn from a known distribution. Based on the severity and contract, the physician chooses the treatment intensity for each ill patient. Treatment intensity, physician productivity, and illness severity jointly determine a patient's health benefits.

Using this model, Section 2.2 shows that relative to a uniform contract, a menu strictly raises welfare if physicians with large private returns to a higher rate also have large social returns. This condition does not hold when physician traits are uncorrelated. Instead, it tends to hold with a positive correlation between cost of effort and altruism, a positive correlation between cost of effort and effective patient severity, or a negative correlation between altruism and effective patient severity. The empirical analog is that a menu is more likely to be efficient when patients with high predicted treatment intensity are registered to physicians with low average treatment intensity relative to patient mix.

2.1 Model

REIMBURSEMENT CONTRACTS. A contract maps treatment intensity m into a physician’s revenue $x(m)$. Motivated by the empirical setting, I focus on contracts with a linear form, also called a two-part tariff: $x(m) = pm + b$. For example, the average physician in my sample receives $p = \$43$ per hour of patient interactions and $b = \$4$ per registered patient per month.¹⁰ Appendix A.3 extends the model to nonlinear contracts.

THE PHYSICIAN. A physician determines treatment intensity m for each registered patient on a panel. Ex-ante, patients are characterized by a distribution of illness severity, $F(\lambda)$. Ex-post, realizations of severity λ are only observed by the physician. The physician also has private information about her type $\theta = \{c, \alpha, \gamma\}$, which is distributed in the population according to $G(\theta)$. Private cost of effort c is an opportunity cost of providing treatment. Altruism α is the weight on utility derived from patient health production relative to utility derived from net income. Both intrinsic and extrinsic forces may motivate physicians to value patient health, e.g., prosociality and reputation. Productivity γ^{-1} measures physician skill in terms of how effectively treatment intensity translates into patient health benefits. A high-productivity physician needs relatively low effort to produce a certain amount of patient health. This notion of productivity is distinct from heterogeneous diagnostic skill (e.g., Abaluck et al., 2016), because here, a low-productivity physician will not treat less intensively due to underdiagnosis.

Before observing realized patient severity, the physician chooses the contract with the highest expected indirect utility: $p_\theta^* = \arg \max E[V(p; \lambda, \theta) \mid \lambda \sim F]$. Following the literature on physician-induced demand, e.g., Ellis and McGuire (1986), indirect utility V is a weighted average of private net income $(p - c)m + b(p)$ and patient health production $h(m, \gamma\lambda)$:

$$V(p; \lambda, \theta) \equiv \max_{m \geq 0} (p - c)m + b(p) + \alpha h(m, \gamma\lambda). \quad (1)$$

After selecting a contract, the physician observes each patient’s severity and chooses a corresponding quantity of treatment: $m^*(p) = \arg \max V(p; \lambda, \theta)$. Incremental treatment will earn additional revenue and influence patient health, but the value does not necessarily outweigh the additional cost of effort.¹¹ At the optimally chosen contract, a physician will be ex-ante indifferent between a small increase in the fee-for-service rate p and the

¹⁰Sections 3.2 and B.2 describe how I construct this aggregate index of reimbursement per hour from data on higher-resolution services like visits, procedures, and diagnostics.

¹¹Appendix A.1 relaxes and tests the assumption of linear cost of effort with a taste for leisure and a constraint on aggregate treatment intensity.

corresponding decrease in the base payment $b(p)$. If $b(p)$ did not decrease in p then all physicians would choose the highest available p .¹² The physician chooses the p_θ^* that sets $-b'(p_\theta^*) = E[m^*(p_\theta^*)] = E[h_m^{-1}(\frac{c-p_\theta^*}{\alpha}; \gamma\lambda)]$.¹³ High rates tend to be selected by physicians who treat intensively, i.e., low cost of effort, high altruism, and low productivity. Since treatment intensity increases in p , an incentive-compatible menu has $b'(p) < 0$ and $b''(p) < 0$.

THE REGULATOR. The regulator observes the population distributions of physician types θ and patient severity λ but not the realizations. The regulator chooses the menu of contracts $b(p)$ to maximize expected patient health production subject to a global budget constraint and each physician's participation constraint.¹⁴ Expenditure, i.e., total payments to physicians, cannot exceed the budget threshold, which incorporates the government's opportunity cost of healthcare spending. Non-health goods and services are also valued and taxation may distort behavior. Participation in the public system is optional, so the expected indirect utility of the physician must stay above a threshold. In the long run, physicians may choose an alternative medical specialty, practice location, or non-healthcare occupation. Physician exit is undesirable because a small number of physicians cannot adequately treat all patients.

The regulator's objective is:

$$\begin{aligned} \max_{b(p)} \quad & \int_{\theta} E[h(m^*(p_\theta^*; \theta), \gamma\lambda; \theta) \mid \lambda \sim F] dG(\theta) \\ \text{s.t.} \quad & \int_{\theta} E[p_\theta^* m^*(p_\theta^*; \theta) + b(p_\theta^*) \mid \lambda \sim F] dG(\theta) \leq \bar{B} \quad [\mu_B, \text{Budget}] \\ & E[V(p_\theta^*; \theta) \mid \lambda \sim F] \geq \bar{v}(\theta), \forall \theta \quad [\mu_{P,\theta}, \text{Participation}] \end{aligned} \tag{2}$$

where μ_B and $\mu_{P,\theta}$ are the shadow costs of expenditure and participation.¹⁵ The social objective partially coincides with the physician objective because of altruism and the participation constraints, but otherwise differs because the regulator is budget-constrained, limiting physician payments. The optimal menu of contracts ("second-best") satisfies the constraints as well as the first-order condition.¹⁶

¹²From the envelope theorem, $\frac{\partial EV}{\partial p} = E[m^*(p)]$, $\frac{\partial EV}{\partial b} = 1$.

¹³The first equality uses $\frac{\partial V}{\partial p} \div \frac{\partial V}{\partial b}$ for a continuous menu $b(p)$. The second equality substitutes using the first-order condition in treatment choice: $p - c + \alpha h_m(m^*, \gamma\lambda) = 0$. Strict concavity of h allows h_m to be invertible in m .

¹⁴Equivalently, the regulator maximizes a weighted sum of expectations over health production, expenditure, and physician indirect utility. In reality, it may be politically difficult to increase healthcare budgets even with a positive aggregate net impact, so I focus on strict constraints until discussing robustness.

¹⁵Privately optimal treatment intensity also depends on patient severity λ which is omitted for readability.

¹⁶Throughout this paper, "optimal" refers to a constrained contract space.

At every potential fee-for-service rate p_k , the first-order condition balances the inframarginal and marginal effects of a small increase in the base payment $b(p_k)$, e.g., by $\epsilon > 0$. To characterize this condition, I take two properties of the menu as given. First, for physicians to sort into multiple contracts, the function $b(p)$ decreases in p . Second, for all $k' \leq k$, $b(p_{k'})$ increases by ϵ to maintain the incentive compatibility of physicians in that range. The inframarginal effect is that ϵ is a lump-sum transfer from the regulator to physicians already choosing $p_{k'}$. The social objective changes by $\mu_{P,\theta} - \mu_B$ for each of those physicians. Since ϵ does not depend on $m^*(p_{k'}^*; \theta)$, these physicians do not change treatment intensity. If the participation constraint is slack, this force pushes towards lower base payments. The marginal effect is that the ϵ -increase in $b(p_k)$ attracts physicians who otherwise would have chosen higher fee-for-service rates ($k' > k$). Each physician who switches will treat less intensively, which lowers both public fee-for-service expenditure and health production. Even before considering inframarginal transfers, some switches can be welfare-decreasing. For these switches, the health loss from lower treatment intensity is greater than the reduction in expenditure, which includes both lower fee-for-service revenue and greater payments. For the optimal menu, the regulator cannot increase any $b(p_k)$ without welfare-decreasing switches and inframarginal transfers outweighing welfare-improving switches.

To benchmark social efficiency, consider the regulator's problem without information asymmetry about physician types θ . In this case, the regulator sets a personalized contract for each physician. The base payment b_θ^{FB} is just high enough for each participation constraint to bind, which implies $\mu_{P,\theta} = \mu_B$. The reimbursement rate p_θ^{FB} induces the efficient level of treatment intensity, $m^*(p_\theta^{FB}; \theta)$ ("first-best"). In expectation, the efficient rate equals marginal health production, weighted by the shadow cost of expenditure:

$$\mu_B p_\theta^{FB} = E[h_m(m^*(p_\theta^{FB}; \theta), \gamma\lambda)].$$

Substituting in the physician's private first-order condition, $p_\theta^{FB} = \frac{c}{1+\alpha\mu_B}$.¹⁷ The efficient reimbursement rate increases with physicians' cost of effort, decreases with altruism, and converges to cost as the budget constraint relaxes. On the other hand, the efficient treatment intensity decreases with the cost of effort and increases with altruism. Productivity does not affect the efficient rate, but lowers the efficient treatment intensity when health returns increase with patient severity.

¹⁷See Appendix C.1 for derivations. Socially efficient treatment intensity is $m^*(p_\theta^{FB}; \theta)$.

2.2 When Is Self-Selection Efficient?

The principal question of this paper is when introducing a choice among contracts (“self-selection”) is socially efficient. In this subsection I present a sufficiency condition and illustrate how efficient self-selection is facilitated by a correlated distribution of cost, altruism, and productivity. The derivations are in Appendix C.1.

The social value of allocating a physician to a marginally higher rate (“social gain”) can be decomposed into two forces, an allocative effect and inframarginal rent:¹⁸

$$\partial_p W_\theta(p) = \underbrace{a_\theta(p)}_{\text{allocative}} - \underbrace{\mu_B s_\theta(p)}_{\text{rents}}.$$

The allocative effect $a_\theta(p)$ is proportional to the gap $(p_\theta^{FB} - p)$ between the physician’s efficient rate and the current rate, scaled by responsiveness $\partial_p m_\theta^*(p)$. The inframarginal rent $\mu_B s_\theta(p)$ is the cost of paying more for treatment that the physician already provides, $s_\theta(p) \equiv E_\lambda[m_\theta^*(p)]$. Putting these together, the social gain is large for physicians far below their efficient rate who treat relatively little. The tension is that physicians with low treatment intensity tend not to choose a higher rate when it is paired with a lower base payment: $\partial_p E_\lambda[V_\theta(p)] = s_\theta(p)$.

Suppose the regulator allows physicians to choose between the best uniform contract (p_0, b_0) and a second contract with a marginally higher rate $p_1 = p_0 + \epsilon$ and lower base payment $b_1 < b_0$. When physicians all have the optimal uniform contract (p_0, b_0) , marginal social value is zero in expectation, $E_\theta[\partial_p W_\theta(p_0)] = 0$, and all participation constraints hold. Optimality of this contract relative to other uniform contracts implies that the allocative effect is positive in expectation: $E_\theta[a_\theta(p_0)] = \mu_B E_\theta[s_\theta(p_0)] > 0$. Equivalently, some physicians have efficient rates above p_0 , which makes a menu potentially valuable. The feasibility of a menu depends on which physicians sort into the higher rate. Physicians with $s_\theta(p) > b_0 - b_1$ will choose the higher rate, so the menu raises welfare when high-treatment physicians tend to have high marginal social value. Equivalently, the average expenditure and health production within the higher-rate contract depend on the set of physicians who choose it. In this way, contract menu design is a selection market like insurance, where allowing choice is efficient when private value and social value are positively correlated across agents (see, e.g., Einav, Finkelstein and Cullen, 2010).

Proposition 1. *There exists a menu that raises social welfare relative to the best uniform*

¹⁸To get this expression, I isolate each physician’s contribution to the regulator’s objective, differentiate with respect to the rate p , and substitute the first-order condition of physician θ .

contract (p_0, b_0) if

$$\text{Cov}_\theta(\partial_p W_\theta(p_0), s_\theta(p_0)) > 0.$$

To a first-order approximation in $\theta = (c, \alpha, \gamma)$ around the average physician, the covariance equals

$$\text{Cov}_\theta(\partial_p W_\theta, s_\theta) \approx \sum_{k \in \{c, \alpha, \gamma\}} \omega_k \text{Var}(\theta_k) + \omega_{c\alpha} \text{Cov}(c, \alpha) + \omega_{c\gamma} \text{Cov}(c, \gamma) + \omega_{\alpha\gamma} \text{Cov}(\alpha, \gamma),$$

with loadings $\omega_c < 0$, $\omega_\alpha < 0$, $\omega_\gamma < 0$, $\omega_{c,\alpha} > 0$, $\omega_{c,\gamma} > 0$, and $\omega_{\alpha,\gamma} < 0$.

Proof sketch. The exact covariance depends on the full joint distribution of θ , so linearize $\partial_p W_\theta$ and s_θ around a physician with average traits, which turns the condition into a bilinear function of the variances and covariances of traits. Indexing θ with j and k , the loading terms are $\omega_k = (a_k - \mu_B s_k) s_k$ and $\omega_{jk} = (a_j - \mu_B s_j) s_k + (a_k - \mu_B s_k) s_j$, where $a_k \equiv \partial_k a_\theta(p_0)$ and s_k is defined similarly. All terms are evaluated at the best uniform rate p_0 and averages of cost, altruism, and productivity. Relabeling the target rate $\tau_\theta \equiv p_\theta^{FB}$, the allocative effect is $\kappa_\theta (\tau_\theta - p_0)$ where κ_θ is a scaling factor. Differentiating $a_\theta(p_0)$ by the product rule $a_k = \kappa_k (\tau_\theta - p_0) + \kappa_\theta \tau_k$. Recall that $\tau_\theta = \frac{c}{1+\alpha\mu_B}$ and $\kappa_\theta = \partial_p m_\theta^*(p_0) \left(\frac{1+\alpha\mu_B}{\alpha} \right)$, so $\tau_c > 0$, $\tau_\alpha < 0$, $\tau_\gamma = 0$, $\kappa_\alpha < 0$, and $\kappa_c = \kappa_\gamma = 0$. Combining terms, $a_c > 0$, $a_\gamma = 0$, and for non-extreme p_0 , $a_\alpha < 0$. Using total differentiation with the physician's first-order condition, $h_m(m_\theta^*, \gamma\lambda) = (c - p_0)/\alpha$: $s_c < 0$, $s_\alpha > 0$, $s_\gamma > 0$. See Appendix C.1 for full details. $\square$

Although necessary, physician heterogeneity ($\text{Var}(\theta) > 0$) is insufficient for a menu to be efficient.¹⁹ Since own-variance terms lower $\text{Cov}_\theta(\partial_p W_\theta, s_\theta)$, incremental heterogeneity weakens the alignment between social ($\partial_p W_\theta$) and private (s_θ) gains. Along any single trait, the physicians who treat more are exactly those for whom a higher rate is least socially valuable. Instead, alignment between social and private gains increases with positive correlation between cost of effort and altruism, positive correlation between cost of effort and inverse productivity, or negative correlation between altruism and inverse productivity. Each of these patterns pairs a trait that raises a physician's efficient rate with one that holds down the treatment intensity through which they would otherwise capture rents, so that the physicians who sort into the higher rate are also those who should face it. Although Proposition 1 addresses the existence of a welfare-improving two-contract menu, the same intuition can help design richer menus of many contracts by characterizing selection between two contracts with adjacent rates.²⁰

¹⁹With no variation in efficient rates ($p_\theta^{FB} = p_0 \forall \theta$), a uniform contract is optimal.

²⁰In a health insurance setting, Chade et al. (2022) “decouple” a related multidimensional menu-design problem into a sequence of two-product comparisons, setting the incremental premium for a marginal

OBSERVABLE COUNTERPARTS. The sufficiency condition is stated in terms of the joint distribution of latent cost, altruism, and productivity, which might not be easily recoverable in another setting. Cross-sectional data on patients' demographics and treatment can help measure the potential for a menu to improve welfare, without estimating a structural model.²¹ Intuitively, if high-severity patients tend to be registered to physicians who provide low treatment relative to that need, then the health returns from a higher rate are large, and those physicians are likely to accept a lower base payment to access the higher rate. To map that intuition to data, use observed patient demographics like age and chronic illness as proxies for patient severity λ . Let $P_j \equiv E_{i \in j}[E[m_{ij} | X_i]]$, where m_{ij} is treatment intensity for patient i with demographics X_i registered to physician j .²² Then, define value-added as $\xi_j \equiv s_j - P_j$, where s_j is the average treatment intensity of physician j , the sample analog of inframarginal rent. A negative correlation between composition-predicted treatment P_j and value-added ξ_j is a favorable indicator that a menu can improve welfare.

The relationship $\text{Cov}(P_j, \xi_j) < 0$ helps meet the sufficiency condition from Proposition 1. Rewriting the condition, $\text{Cov}(a_j, s_j) > \mu_B \text{Var}(s_j)$, i.e., less dispersion in rents means a lower threshold for the covariance between allocative effects and rents. Dispersion can be decomposed as

$$\text{Var}_j(s_j) = \text{Var}_j(P_j) + \text{Var}_j(\xi_j) + 2 \text{Cov}(P_j, \xi_j),$$

where a negative $\text{Cov}(P_j, \xi_j)$ lowers the threshold. Moreover, $\text{Cov}(P_j, \xi_j) < 0$ can imply a greater $\text{Cov}(a_j, s_j)$ to the extent that cost and altruism determine value-added. Physicians with high average patient severity $\bar{\lambda}$ and high cost c or low altruism α have both relatively high treatment intensity ($\bar{\lambda}$) and large allocative effects (c, α). That combination corresponds to the covariance loadings in Proposition 1 once the model is extended to include $\bar{\lambda}$ as a fourth dimension of physician heterogeneity: $\omega_{\bar{\lambda}} < 0$, $\omega_{c\bar{\lambda}} > 0$, $\omega_{\alpha\bar{\lambda}} < 0$.²³ In addition to $\text{Cov}(P_j, \xi_j) < 0$, when panel variation is available, another favorable indicator is high average responsiveness of treatment to the fee-for-service rate, which scales the allocative effect. On the other hand, if an increase in the rate results in worse health outcomes, i.e., negative marginal health production, then the status quo rate likely exceeds the best uniform rate.

increase in coverage. In that setting, decoupling requires quasiconcave household utility in coverage. In my application, the optimal menu satisfies a related condition: each physician's expected indirect utility is single-peaked in the reimbursement rate among traded contracts.

²¹Estimating latent physician traits requires panel variation in the fee-for-service rate.

²² P_j can be estimated using only the patients of other physicians, so it does not reflect physician j 's own behavior.

²³Signs are identical to γ loadings except for the added $\omega_{\gamma\bar{\lambda}} = -2\mu_B s_\gamma s_{\bar{\lambda}} < 0$. If $\text{Cov}(P_j, \xi_j) < 0$ only because $\text{Cov}(\gamma, \bar{\lambda}) < 0$, then it is not informative about $\text{Cov}(a_j, s_j)$.

3 Empirical Setting

The theoretical framework characterizes how the distribution of physician types determines whether a menu of contracts increases welfare relative to a uniform contract. Below, I extend the framework to estimate such a distribution, derive the optimal menu, and measure its impacts. In this section, I first establish several necessary assumptions in the setting of Norwegian primary care. I present institutional details in Section 3.1 which support the assumption that the focal variation in treatment intensity is driven by physician heterogeneity and contracts rather than patient composition. In Section 3.2 I detail the construction of a balanced estimation sample of patients that further removes potentially confounding variation. In Section 3.3 I introduce reduced-form evidence consistent with physician heterogeneity in cost, altruism, and productivity, which satisfies a necessary condition for a welfare-improving menu.

3.1 Institutional Setting

In Norway, each practicing primary care physician can increase their reimbursement by becoming certified as a general practitioner. In 2023, physicians without the certificate received \$33 for a basic consultation and certified physicians received \$44. As a result, with no changes to treatment intensity, a newly certified physician would suddenly earn 24 percent more fee-for-service revenue.²⁴ Crucially for causal inference, certification does not formally change a physician’s patient pool, treatment options, or responsibilities. Physicians become eligible for the certificate by completing two years of additional part-time training and also having four years of full-time practice experience. Training includes both coursework and small-group meetings with other physicians, guided by national learning objectives.²⁵ Once the training is completed, physicians can apply for the certificate, which they typically receive after several months. Supplementary payments begin around that time and continue for five years. The lag between the end of training and the start of supplementary payments helps attribute the short-run change in treatment intensity to financial incentives rather than training-related time constraints. During the sample period, 80 percent of physicians received this certificate during their careers.²⁶

Apart from certification, physicians face nationally uniform reimbursement incentives.

²⁴24 percent reflects the average change in total fee-for-service revenue within the estimation sample.

²⁵In 2019, physicians needed to meet 88 learning objectives. For example, Objective #18 is to understand challenges with over- and under-treatment.

²⁶In March 2017, it became mandatory for most primary care physicians to start training toward certification. In March 2019, municipalities became responsible for facilitating supervised hours requirements and subsidizing part of the costs.

On average, physicians receive 70 percent of their revenue from fee-for-service payments, at rates listed in a national administrative schedule.²⁷ For example, in 2021, physicians received \$17 for an E-consultation, made up of \$16 from national health insurance and \$1 from a patient copay (Den Norske Legeforening, 2022).²⁸ In 2023, the schedule included 189 reimbursement codes, covering broad categories of physician services. The most commonly billed codes cover unspecified time spent with patients, rather than a specific procedure or diagnostic, highlighting the importance of physicians’ discretion in choosing treatment intensity.²⁹ The other 30 percent of revenue comes from base payments of approximately \$4 per registered patient per month. Both fee-for-service rates and base payments are negotiated annually between the regulator and the physicians’ union. If prices were instead negotiated individually between physicians and payers, as is common in the United States, it would be difficult to attribute variation in treatment intensity to reimbursement rates rather than physician skill or patient composition.

Within the scope of these national reimbursement agreements, physicians contract directly with municipalities. Among other details, these contracts stipulate the maximum number of registered patients and opening hours. Each physician agrees to meet the primary care treatment needs of between 500 and 2500 registered patients. National guidance states that physicians must be accessible to registered patients within contracted opening hours and that patients should not wait more than five days for a consultation in most circumstances (Lovdata, 2017). If physicians are unavailable, registered patients may seek treatment from stand-alone urgent care centers. Physicians provide consultations about symptoms, diagnostic tests, and general medical procedures to registered patients. They also approve sick leave and gate-keep referrals for scheduled visits to specialists and hospitals.

Patients often choose to remain with their registered physician for years at a time. One contributing factor is the centralized registration system, which allows patients to request a new physician twice per year. Patients can choose any physician who has fewer patients than the contracted maximum. The choice set changes infrequently due to the national licensing system, which fixes the total number of local physicians in the short term. Since patients rarely switch, I can construct a large balanced panel of patients for the estimation sample that remains nationally representative.³⁰

²⁷As of 2016, over 95 percent of physicians face this mixed contract. The remainder are fixed-salary employees of municipalities with no fee-for-service reimbursement.

²⁸Once a patient reaches an annual individual cap on copayments, the public insurer funds the entire \$17.

²⁹In the United States, most claims for primary care consultations also include one of a small number of procedure codes.

³⁰Table A.1 shows that the sample and full population have similar averages of patient age, gender, chronic illness, and disability.

3.2 Data

The estimation sample is a balanced panel of patients who are registered to certified physicians in the six months before and after certification (a “spell”).³¹ I focus on short-term variation and fix the composition of patients to attribute any sudden change in treatment intensity to the sudden change in the fee-for-service rate. I construct the sample using restricted administrative records on registration, individual demographics, and healthcare reimbursement, which are maintained by Statistics Norway, the Directorate of Health, and the Institute of Public Health.³² These records nearly span the universe of Norway’s residents and primary care physicians from 2008 to 2017.

The estimation sample excludes potentially confounding variation. First, each physician must only practice in one location during the entire period and each patient must be registered for the entire period. Second, both the physician and patient must have identification numbers to attribute treatment intensity to a particular physician of interest, which excludes recent migrants. I measure utilization at urgent care centers and second-opinion visits as a separate outcome. Third, each physician must provide some treatment during every month of the spell to exclude irregular variation that arises from the physician’s absence, e.g., an anticipatory effect or temporary replacement physician. Fourth, by restricting to short-term variation, I exclude the potential effects of training on utilization. Figure A.2 shows that utilization falls 8-24 months before certification, possibly due to training. In robustness analyses, I compare the estimation sample to a similarly defined control sample that includes patients whose physicians do not experience sudden changes in reimbursement.³³

I construct measures of treatment intensity and marginal reimbursement rates that aggregate over the particular types of services provided. I construct treatment intensity m to roughly correspond to hours of treatment per patient-month. This simulated hours measure equals patient-month fee-for-service revenue divided by p_{kt} , where p_{kt} is a “simulated wage” equal to the fee-for-service revenue per hour a physician would receive for providing the average bundle of services to a patient of type k in month t . Going forward, I refer to p_{kt} as a fee-for-service rate. I group patients with similar characteristics into ten types, and for

³¹I classify the first month a physician is certified based on when they first receive a supplementary payment, including reimbursement codes 2dd, 2dk, 6ad, 11dd, 11min, and 14d, which is generally consistent with the certification date.

³²See Appendix B.1 for additional details on data sources: Norwegian Directorate of Health (2020*a,b*); Norwegian Institute of Public Health (2020); Statistics Norway (SSB).

³³To accommodate computer memory constraints, I use a 10-percent random subsample of physicians who never receive the certification supplement during the study period. I randomly select a 13-month spell that meets the same conditions as the main estimation sample, except for certification. Spells prior to certification are also safe comparisons, but I exclude these from the control sample to be conservative when analyzing selection into certification.

Table 1: Registered Patient Summary Statistics

	Control Sample	Estimation Sample					
	Mean	Mean	Std. Dev.	% > 0	10th	50th	90th
Patient Characteristics							
Reimbursement	8.59	8.33	25.49	20.74	0.00	0.00	30.92
Simulated Hourly Rate	43.82	43.76	6.86	100.00	32.38	45.49	50.95
Simulated Hours	0.19	0.18	0.56	20.74	0.00	0.00	0.68
Base Payment	4.03	4.01	0.11	100.00	3.84	4.02	4.13
Age	40.54	37.57	22.78	100.00	6.67	36.58	69.00
Chronic Illness	0.23	0.21	0.41	21.03	0.00	0.00	1.00
Months Registered	43.89	40.93	32.32	98.99	6.00	36.00	84.00
Physician Characteristics							
Max Enrollment	1268.60	1273.48	293.21	100.00	900.00	1220.00	1600.00
Physician Hours/Week	28.36	26.56	9.44	100.00	13.13	27.33	37.27
Physician Age	42.87	40.23	5.92	100.00	34.08	38.83	48.67
Patients Age 60+	0.23	0.19	0.10	100.00	0.07	0.18	0.32
Patients with Chronic Illness	0.23	0.21	0.06	100.00	0.14	0.20	0.29
Patients	131800	643363					
Physicians	136	619					

Notes: Summary statistics reflect registered patients' monthly totals six months before certification (or a randomly selected month for patients in the control sample). % > 0 indicates the share of patients with a strictly positive value. Other columns reflect the mean, standard deviation, and 10th, 50th, and 90th percentiles. Monetary measures are in USD. Physician Characteristics are also averaged across patients. The last two Physician Characteristics reflect shares of registered patients.

each type, I use all Norwegian patients to calculate the average bundle of services received and the average hours required to provide that bundle.³⁴ I inflate all money-metric variables by Norway's monthly all-goods-and-services CPI to January 2023 USD.

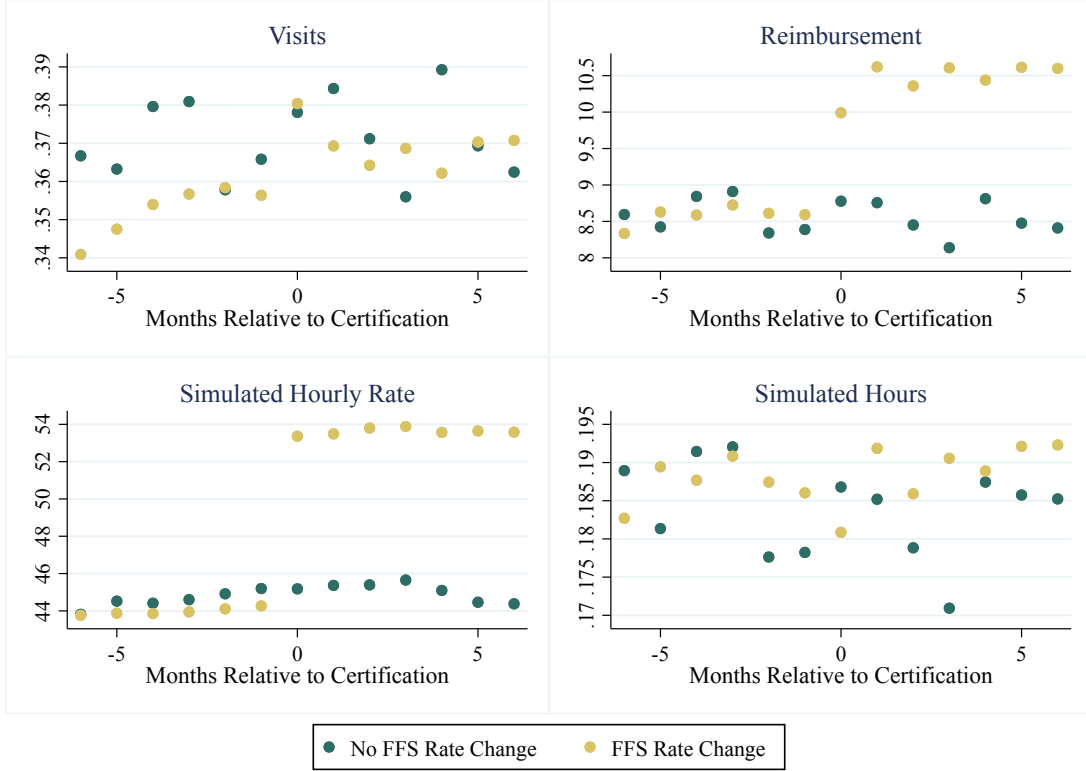
The final estimation sample is approximately representative, and it includes 619 unique physicians and 643,363 patient-spells (13 months each).³⁵ Table 1 describes the distribution of selected characteristics and outcomes six months before certification, and three facts stand out.³⁶ First, most patients do not visit their physician during a typical month. Second, the average physician spends 27 hours per week with registered patients (90th percentile = 37) suggesting that with strong enough incentives, physicians could increase treatment intensity. Third, there is meaningful heterogeneity across physicians for proxies of mean

³⁴See Appendix B.2 for additional details on constructing measures. For example, hours reflect time spent in encounters with registered patients and not work like administrative tasks. Table A.2 shows average characteristics and sample share separately for each patient type, including the simulated wage.

³⁵When estimating the structural model, I split this sample into three parts and use the best-fitting set of estimates.

³⁶See Table A.1 for comparisons to the Norwegian population: patients have similar characteristics, and certifying physicians are more often young and female.

Figure 1: Trends Relative to Certification



Notes: These plots show averages of outcomes across patient-months in the estimation and control samples in each month relative to certification. Each sample is a balanced panel of patients, and in the estimation sample, Month 0 is the first month in which the registered physician received a certification supplement. I include registered patients' visits and fee-for-service reimbursement for the focal certified physician. Simulated hours equals monthly reimbursement divided by the simulated hourly rate, with construction details in Appendix B.2.

patient severity like average age and chronic illness. To separately identify the productivity and cost of effort parameters, it is important to observe variation both within and across physicians in characteristics that shift patient severity.

Treatment intensity varies systematically with the fee-for-service rate, and short-run changes are persistent. Figure 1 plots the trend in raw means, showing that visits, total reimbursement, and simulated hours all increase suddenly after certification in the estimation sample but not the control sample. Unlike treatment intensity, trends in the number and composition of registered patients do not change with certification (See Figure A.1). These plots and most subsequent analyses reflect short-run variation around the sudden change in incentives which usually occurs months after physicians complete the prerequisite training. Short-run variation might obscure differences in long-run trends between certified and non-certified physicians that limit validity. For example, physicians who pursue certification

might also make cost-reducing investments, or training might have delayed effects. Mitigating these concerns, Figure A.2 shows that even over five years, certification corresponds to a sudden and persistent increase in related measures of treatment intensity. Raw means suggest that the effects of certification might be overstated if using a longer time horizon, because treatment intensity dips during months which likely correspond to the middle of training.

3.3 Stylized Facts

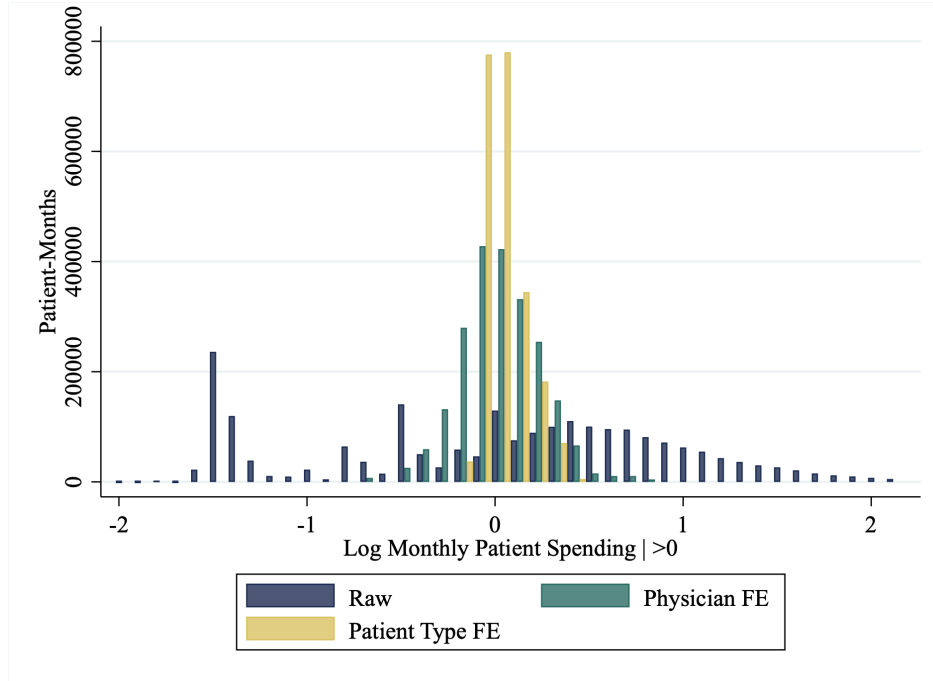
A necessary condition for physician self-selection is variation in physician types. I show novel reduced-form evidence consistent with heterogeneity in physicians’ cost, altruism, and productivity. First, I show descriptively that observably similar patients receive more treatment at some physicians than at others, driving a large share of variation in treatment intensity. Second, I exploit quasi-random patient assignment to estimate the heterogeneous causal effects of physicians on treatment and adverse outcomes, consistent with variation in cost and productivity. Third, with a stacked difference-in-differences model, I show that treatment intensity increases in the fee-for-service rate across a range of measures, highlighting the role of altruism. Heterogeneity in this effect implies dispersion in altruism.

COST AND PRODUCTIVITY. Figure 2 shows the persistent variation across physicians in how intensively they treat observably similar patients. To make this comparison, I regress log reimbursement on fixed effects for each physician and 108 bins of patients with similar observed characteristics, as well as other controls.³⁷ Reimbursement per patient-month is approximately log-normally distributed with significant dispersion, while variation across patients with different observed characteristics (e.g., age, gender, chronic diagnoses) is relatively small. The limited dispersion across patients’ observed characteristics implies that the regulator can only weakly predict patients’ underlying treatment need and must generally defer to physicians’ judgment about the appropriate level of treatment intensity. Physician fixed effects are more dispersed, highlighting the large role of physicians in treatment intensity, similar to evidence from moves, e.g., Badinski et al. (2023).

These physician fixed effects should not be interpreted causally if, for example, patients with high unobserved severity systematically seek care from physicians with high expected treatment intensity. Fortunately, conditionally random patient assignment in Norway allows me to recover plausibly causal estimates of assignment to each physician (“assignment ef-

³⁷I regress log reimbursement on an indicator for post-utilization, physician fixed effects, high-resolution patient-type fixed effects (combinations of age bins, primary diagnosis, gender, and an indicator for lagged hospitalization), a time trend, and a quadratic function of patient age, among patient-months with positive reimbursement.

Figure 2: Decomposition of Treatment Intensity



Notes: This histogram shows the plot of log reimbursement for patient-months in the estimation sample with any utilization, as well as fixed effects from a regression of that outcome on an indicator for post-certification, physician fixed effects, high-resolution fixed effects for patients with similar observed characteristics (combinations of age bins, primary diagnosis, gender, and an indicator for lagged hospitalization), and a quadratic function of patient age.

fects”) on subsequent log treatment intensity, following the approach in Ginja et al. (2025).³⁸ As shown in Figure A.4, there is substantial dispersion in these physician effects even after shrinking effects to account for estimation error, reinforcing the importance of persistent physician heterogeneity. Limited patient selection is consistent with evidence from Norway that a patient’s choice of physician is uncorrelated with the physician’s effect on mortality (Ginja et al., 2025). In Norway and other settings, patients tend to respond to public measures of quality like star ratings (Bensnes and Huitfeldt, 2021; Vatter, 2025; Brown et al., 2023; Chartock, 2025). By contrast, treatment intensity does not appear to drive patient switching (Iversen and Lurås, 2011).

Continuing to use random patient assignment, I estimate effects of individual physicians on related outcomes to distinguish cost and productivity as drivers of persistent physician heterogeneity. In the model, low-productivity physicians treat patients multiplicatively more

³⁸When one physician exits, the municipality reassigns remaining patients to nearby available physicians, and the assignment is conditionally random. This variation exists for a subset of physicians. The research design compares patients of the same exiting physician who are assigned to different nearby physicians to recover those nearby physicians’ assignment effects, controlling for the exiting physician, year, and nearby physician’s municipality and availability. I shrink all physician assignment effects using Empirical Bayes.

– leading to variation in assignment effects on log reimbursement – while low-cost physicians treat patients additively more – leading to variation in levels of reimbursement. Figure A.4 shows significant variation among both sets of assignment effects. For example, moving from the 10th to 90th percentile of physician treatment intensity corresponds to 1.19 additional visits each month over a patient mean of 0.34. In the model, when two physicians provide identical treatment to identical patients, patient health can still differ due to productivity. Likewise, I document variation in physician assignment effects on avoidable hospitalizations, and this variation is only weakly correlated with physician assignment effects on treatment intensity. The dispersion in cost and productivity is consistent with evidence from other natural experiments that physicians vary in resource use and quality measures, such as avoiding hospital readmission (Doyle, Ewer and Wagner, 2010; Gowrisankaran, Joiner and Léger, 2023; Chan, Gentzkow and Yu, 2022; Chan and Chen, 2026; Kwon, 2023).

ALTRUISM. Altruism is identified by how physicians’ choice of treatment intensity responds to the reimbursement rate. Intuitively, relatively altruistic physicians have less scope to change treatment intensity when the reimbursement rate changes. At any reimbursement rate, these physicians sacrifice profit to provide greater health production.³⁹ To evaluate the effect of higher reimbursement from certification on treatment intensity, I estimate the following stacked difference-in-differences regression:

$$Y_{ijt} = \beta_1 Post_{jt} \times Certified_j + \beta_x \mathbf{X}_{jt} + \gamma_i + \gamma_{y(t)} + \gamma_{m(t)} + \epsilon_{ijt} \quad (3)$$

where Y_{ijt} is the outcome of interest for patient i of physician j in month t . $Post_{jt}$ is an indicator for months in which physicians receive certification supplements, $Certified_j$ indicates the main estimation sample of certified physicians rather than randomly selected non-certified physicians. β_1 is the coefficient of interest, and $\gamma_i, \gamma_{y(t)}, \gamma_{m(t)}$ are fixed effects for patient, year, and calendar month. X_{jt} controls for the number and composition of registered patients.⁴⁰

The main threat to identification is the possibility of certification changing something other than financial incentives. Given institutional details, it seems unlikely that existing patients of certified physicians systematically and suddenly need more treatment in the six months after certification, beyond the variation captured by patient fixed effects and time fixed effects. First, physicians are not suddenly eligible to provide more expensive services.

³⁹For any health production function with fixed concavity, the responsiveness of treatment intensity to the fee-for-service rate, $\frac{\partial m}{\partial p}$, is proportional to inverse altruism, $\frac{1}{\alpha}$, among patients with positive treatment intensity.

⁴⁰Composition include the share of patients age 20-39, age 40-59, age 60+, female, and with asthma, cancer, COPD, CVD, or diabetes.

Second, as shown in Figure 2, future treatment need is difficult to anticipate, so physicians likely have little scope or incentive to strategically time their application for certification after completing the training. Alternative explanations are generally incompatible with Figure 1, which shows that average reimbursement does not trend differently for certified versus non-certified physicians in the months before certification.⁴¹

Table 2: Main Effects of Certification on Treatment Intensity

	Post $\times$ Certified		Mean (Pre)	R ²	Obs.
Visits	0.015***	(0.001)	0.355	0.401	9,301,956
Reimbursement	2.093***	(0.106)	8.581	0.213	9,301,956
Simulated Hours	0.006**	(0.002)	0.187	0.186	9,301,956
Procedures	-0.001	(0.001)	0.071	0.237	9,301,956
Diagnostics	0.009***	(0.002)	0.229	0.266	9,301,956
Extra Time Codes	0.002***	(0.001)	0.086	0.230	9,301,956
Other Reimbursement	-0.303***	(0.076)	2.486	0.099	9,301,956
Specialist Reimbursement	0.245	(0.310)	19.702	0.190	9,301,956
Acute Hospitalizations	-0.000	(0.000)	0.019	0.153	9,301,956

Notes: This table shows estimates of the coefficient on currently being certified and thus eligible for a higher rate (See equation 3). The unit of analysis is a patient-month and the sample includes registered patients in the six months before and after a physician becomes certified. The sample also includes non-certified physicians indexed to a placebo month. Standard errors are clustered by physician. Visits includes any in-person encounter. Reimbursement indicates fee-for-service revenue. Simulated Hours is reimbursement divided by a price index as described in Section 3.2. Procedures, Diagnostics, and Extra Time Codes are counts of reimbursement codes grouped by the chapter of the reimbursement code. These categories are mutually exclusive but not exhaustive. Other Reimbursement includes treatment by any primary care physician other than the registered one, e.g., at community health clinics. Specialist Reimbursement includes all non-primary physician care eligible for public reimbursement. Acute Hospitalizations are unscheduled with admission within six hours. Mean (Pre) is an average of patient-months in the six months before certification, using only the certified-physician sample. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level.

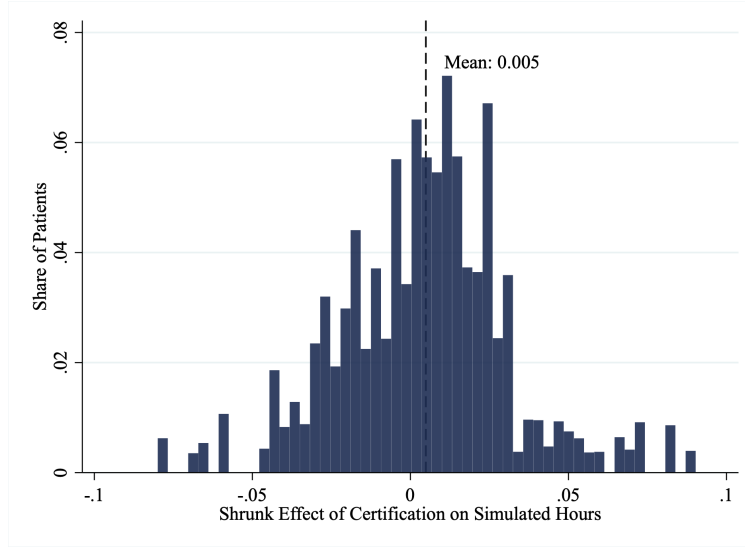
Table 2 shows that higher reimbursement rates result in greater treatment intensity. I observe precise increases both in visits, which are directly incentivized, and other measures of treatment intensity, which suggests complementarity between visits and in-visit services.⁴² Simulated hours, which combines all categories of treatment, increase by approximately 3 percent of the pre-certification mean. Relative increases are similar for sub-categories of reimbursement codes like diagnostics and extra time per visit. Notably, increased treatment

⁴¹See Section 3.2 for discussion of the long-run variation shown in Figures A.2.

⁴²Brekke et al. (2017) perform a similar analysis, finding a comparable effect on visits but no evidence of effects on treatment intensity per visit. The difference might be due to lower power from the narrower sample, confounding effects of changing patient composition from the underlying unbalanced patient panel, or confounding time-specific shocks from the lack of a comparison group like non-certified physicians.

intensity provided by the registered physician coincides with small decreases in primary care from other physicians. The counterfactuals below focus on the treatment intensity of registered physicians and might overstate incremental expenditure from higher rates relative to this substitution effect.⁴³ I do not find evidence that certification immediately affects specialist treatment or acute hospitalizations.

Figure 3: Distribution of Physician-Level Effect of Certification on Simulated Hours



Notes: This histogram shows estimates of β_{1j} from equation 3 where the effect of certification is allowed to vary by certified physician. I shrink estimates to the mean using Empirical Bayes. Frequencies are weighted by the number of patients. Estimates are based on a subsample of spells starting 2010-2012.

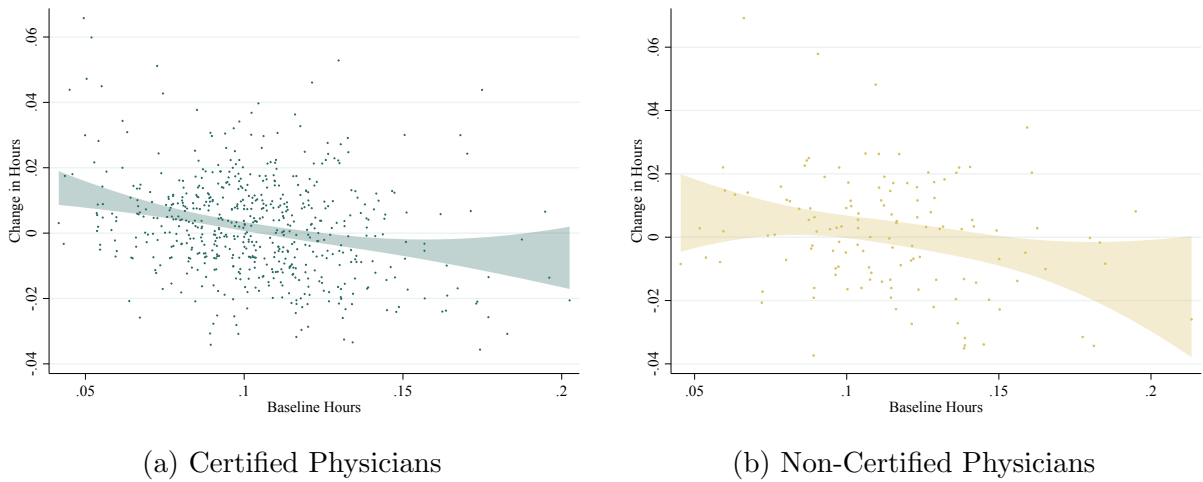
Consistent with dispersion in physicians' altruism, I find heterogeneity in the effect of certification on treatment intensity. I extend the difference-in-differences analysis to include a post-certification indicator for each physician. Figure 3 is a plot of the physician-specific estimates after adjusting for error. Although the average physician increases treatment intensity post-certification, there is meaningful heterogeneity including precise negative estimates, motivating the test for income effects in Section 6.2. Estimates do not correlate precisely with physicians' observed characteristics like employment history or the maximum number of patients. However, responsiveness in simulated hours do correlate with responsiveness across alternative measures of treatment intensity (Figure A.3). Dispersion in altruism is consistent with experimental evidence of heterogeneity (Godager and Wiesen, 2013; Hennig-Schmidt, Selten and Wiesen, 2011). To interpret estimated elasticities exclusively as altruism, physicians must not vary in their ability to increase treatment intensity. In Section 6.2 I discuss several tests of this assumption. For example, in Figure A.13 I present descriptive evi-

⁴³Changes to health production might be understated if registered physicians do not fully internalize substitution with other providers.

dence that capacity constraints do not bind in this setting and in Figure A.12 I show that high-altruism and low-altruism physicians respond similarly after observed shocks to patient health.

CORRELATION. Dispersion in physicians' cost of effort, productivity, and altruism satisfy a necessary condition for physician self-selection. However, to separate physicians across contracts, these dimensions of heterogeneity should also be correlated. In Section 2.2, I illustrate how physicians with high efficient reimbursement rates must have relatively high willingness-to-pay for higher rates. Before estimating the correlation structure for physician types in the next section, I check for a consistent pattern in the raw data. Figure 4 shows that when the reimbursement rate increases, physicians with large increases in treatment hours (i.e., from low altruism) tend to initially provide low treatment intensity (e.g., from high cost or low productivity). This pattern does not appear among physicians without reimbursement-rate variation, so it is unlikely to only reflect regression to the mean or idiosyncratic noise. Relative to Section 2.2, the negative correlation is consistent with both unfavorable ($\text{Cov}(c, \alpha) < 0$) and favorable ($\text{Cov}(\alpha, \gamma) < 0$) indicators for efficient self-selection. This ambiguity highlights the need to fully estimate the latent distribution of physicians' cost, altruism, and productivity.

Figure 4: Raw Data Consistent with Correlated Altruism

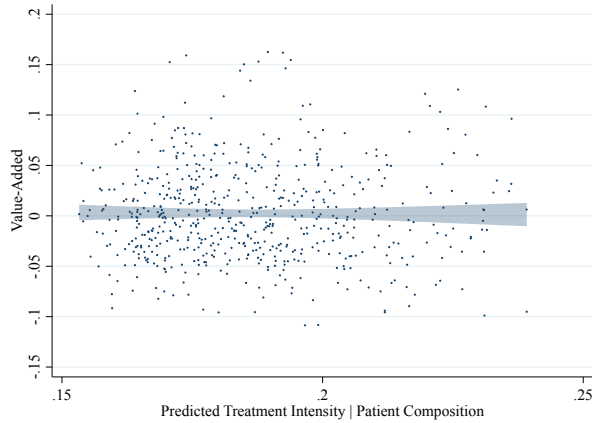


Notes: These plots show the correlation between pre-certification treatment intensity and the change in treatment intensity (post-certification relative to pre). Each point is a physician, and the pooled sample truncates the lowest and highest percentile of both axes. I calculate the average hours of treatment per registered patient in the six months before certification and the six months after certification. The placebo certification date is randomly selected for the control sample of non-certified physicians. The shaded region indicates a 95 percent confidence interval for a quadratic prediction.

PATIENT COMPOSITION. I do not find evidence that high-need patients are registered to certified physicians who provide low treatment relative to that need. This test is the

observable counterpart to the sufficiency condition from Proposition 1. The implication is that variation in patient severity across physicians does not align private and social gains towards efficient self-selection. A welfare-increasing menu is still possible if correlated cost, altruism, and productivity align allocative effects with inframarginal rents. I predict each physician’s average treatment intensity using patients’ age, gender, and diagnoses, based on observably similar patients at other physicians. Figure 5 shows the negative correlation between physicians’ P_j and value-added, $\xi_j \equiv s_j - P_j$, where s_j is the observed average treatment intensity. The corresponding regression coefficient, -0.024 (0.105), is negative but small and imprecise. It would need to be at least 0.5 for physician value-added to fully offset the direct impact of patient severity on inframarginal rents.

Figure 5: Low-Treatment Physicians Do Not Tend to Treat High-Need Patients



Notes: Both axes are measured as simulated hours per patient-month. Each point is a certified physician, truncating the lowest and highest percentile. Predicted treatment intensity (P_j) is a physician-level weighted average of patient-level predicted treatment intensity, using the same demographic-based patient bins as Figure 2. Patient-level predicted treatment intensity is calculated using only non-focal physicians. Value-added ξ_j is the residual: observed average treatment intensity per physician minus P_j .

4 Empirical Model

I estimate the joint distribution of physician heterogeneity to predict behavior under counterfactual menus and determine whether introducing a menu would increase efficiency relative to a uniform contract. In this section I review additional assumptions to support estimation, including intuition for which patterns in the data help to recover each parameter.

4.1 Identification Intuition

The key condition supporting identification of physician traits θ is that the data include orthogonal within-physician variation in the fee-for-service rate p and observed patient demographics X . Any unobserved variables that affect patient severity λ must be uncorrelated with p , X , and θ .⁴⁴ The identifying variation in p and X separately shifts each side of the physician's first-order condition, $h_m(m_{ijt}, \gamma_j \lambda_{ijt}) = \frac{c_j - p_{jt}}{\alpha_j}$, where i indexes patients, j indexes physicians, and t indexes time. Rate p is the financial return to treatment intensity, and patient demographics X shift the health return to treatment intensity, $h_m(m, \gamma \lambda)$, via patient severity, λ . Given physician j and rate p , $\frac{c_j - p_{jt}}{\alpha_j}$ is fixed across patients, so the conditional variance in treatment intensity $\text{Var}(m_{ijt} | j, p)$ is driven by the conditional variance in effective severity $\gamma_j \lambda_{ijt}$. Within effective severity, differences in expected patient severity $E[\lambda | X]$ across demographics are identified by the corresponding differences in treatment intensity, conditional on the physician and rate, with dispersion in residuals pinning down the variance of severity. Differences in inverse productivity γ are identified by differences between physicians in the conditional variance that is not explained by differences in patient demographics X_{it} . The first-order condition also implies that altruism α is identified by the physician-specific effect of the rate on treatment intensity, $\partial_p m^*$.⁴⁵ Differences in cost c are identified by differences between physicians in average treatment intensity, given altruism, productivity, patient demographics, and the rate. To separately identify the level of productivity from the level of severity, either the productivity of a reference physician or the severity of a reference patient must be normalized. To separately identify the level of severity from average cost, severity must be a nonlinear function of its determinants.

In Figure A.5, I show stylized visual examples of the identifying variation. If two physicians and their patients are otherwise identical, then dispersion in treatment intensity across patients is greater for a physician with lower productivity (greater γ). Average treatment intensity of a relatively altruistic physician increases by a relatively small amount after the fee-for-service rate increases. Average treatment intensity is persistently lower for a relatively high-cost physician, across both rates and patient demographics.

4.2 Parameterization

I estimate the distributions of physician heterogeneity and patient illness severity by maximizing the likelihood of observed treatment intensity. Privately optimal treatment intensity sets marginal net income equal to marginal health production scaled by altruism. To gener-

⁴⁴In Sections 3.3 and 6.2 I discuss evidence supporting this assumption.

⁴⁵Differentiate with respect to p , then $h_{mm}(m^*, \gamma \lambda) \partial_p m^* = -\frac{1}{\alpha}$, where $h_{mm} < 0$.

ate a likelihood, I make two parametric assumptions that I later relax in Section 6.2. First, since economies of scale are unlikely in this setting, I continue to assume that costs increase linearly in treatment intensity: $c(m) = cm$.⁴⁶ Second, health production is quadratic in the distance between treatment intensity and patient severity scaled by productivity: $h(m, \lambda; \gamma) = H - \frac{1}{2}(m - \gamma\lambda)^2$. Quadratic functional forms are common in the insurance literature to model households' valuation of treatment intensity, e.g., Cardon and Hendel (2001), Einav et al. (2013), and Marone and Sabety (2022). Given these assumptions, privately optimal treatment intensity takes the form:

$$m^*(p, \lambda, F) = \max\{0, \frac{p - c}{\alpha} + \gamma\lambda\}. \quad (4)$$

Gaynor, Mehta and Richards-Shubik (2023) use a special case of this parameterization where γ is constant across physicians and λ is a deterministic function of patient characteristics.

The final step is to solve for the model residual, the unobserved component of patient severity. I parameterize the distribution of severity as a two-stage process. Conditional on being positive, severity is distributed log-normal to help rationalize the skewed distribution of observed treatment intensity. Average log severity varies with observed characteristics, satisfying the nonlinearity condition for identification in Section 4.1: $(\ln \lambda \mid \lambda > 0) \sim N(\beta_\lambda X_\lambda, \sigma_\lambda)$.⁴⁷ I parameterize the probability that severity is positive as $Pr(\lambda > 0) = \frac{\exp d_0 + d_1 \beta_\lambda X_\lambda}{1 + \exp d_0 + d_1 \beta_\lambda X_\lambda}$. This step helps rationalize why patients often have zero treatment intensity, similar to Ho and Lee (2023). Appendix C.2 presents the full expression of the conditional likelihood.

4.3 Estimation

To recover parameters of the model, I maximize the likelihood of observed treatment intensity for patients of certified physicians in the six months before and after a change in the fee-for-service rate from certification: $l(m \mid \theta_i, p, F)$.⁴⁸ Parameters include the conditional

⁴⁶For example, the regulator dissuades a large number of patients per physician by approving the entry of each new practice. Similarly, the maximum number of patients per physician can be up to 2500 but most physicians choose a much lower maximum. I exclude the small number of physicians who share a workload with other physicians.

⁴⁷These characteristics include fixed effects for each of the 10 observed patient types, fixed effects for calendar months, normalized lagged treatment intensity, an indicator for zero lagged treatment intensity, indicators for cancer, diabetes, COPD, Asthma, and CVD, indicators for 1 or 2+ of these chronic illnesses, indicators for female and disability receipt, percentile of income as of 2016, indicators for 1 or 2+ acute hospital visit in the last 6 months, and indicator for registering with the current physician in the last 6 months and a scaled time trend.

⁴⁸I use L-BFGS-B with the Python module JAX to calculate the analytic gradients of the log-likelihood objective. The box constraints are that cost, altruism and productivity are strictly positive and that

means and variance of patient severity $F(\lambda)$, and each certified physician’s marginal cost c , altruism α , and productivity γ^{-1} . Estimated parameters are sometimes simple transformations of model parameters.⁴⁹ Since inverse productivity γ scales patient severity λ , a high population average of λ is not separately identified from high average if γ , so I fix the intercept of log severity $\beta_{\lambda,0}$ at zero.⁵⁰ To accommodate computer resource constraints, I separately estimate parameters for three subsamples defined by the year of certification: 2008-2010, 2011-2013, and 2014-2016. I use the 2011-2013 subsample for counterfactuals because parameter estimates best predict treatment hours. To estimate parameters for the control sample of non-certified physicians, I follow the same procedure except that altruism α is fixed. Since the control sample lacks identifying variation in the fee-for-service rate, I predict altruism using observed physician characteristics, based on correlations in the certified-physician sample (See Table 3).

5 Estimates

Parameter estimates are sensible and fit the data, accurately predicting treatment intensity both in- and out-of-sample. To assess the model fit, I first plot observed treatment intensity against predicted values. Figure 6 shows a correlation of nearly 1 for both the estimation sample and a control sample of never-certified physicians.⁵¹ Estimates predict treatment intensity well both across physicians and over time for particular physicians. Table A.4 shows corresponding regressions: the coefficient on predicted treatment intensity is approximately 1, even when including physician fixed effects in columns (3) and (5). Column (5) shows that, conditional on estimates, patient demographics explain little remaining variation in treatment intensity.⁵²

The correlation between estimated cost, altruism, and productivity reinforces the potential for efficient self-selection. Figure A.6 shows the joint density of physician heterogeneity. Relatively productive physicians tend to have low cost of effort and high altruism, which

marginal cost is no more than ten times as large as marginal revenue.

⁴⁹ c is a multiple of the fee-for-service rate six months before certification, α is scaled by 1000, and σ_λ is exponentiated. The transformation of c implies that marginal cost varies across patients of the same physician.

⁵⁰With this normalization, I assume that a young long-term low-income male patient with no major diagnoses or lagged utilization in the first month of the sample has diminishing returns to treatment after $\hat{\gamma}_j$ hours per month, where $\hat{\gamma}$ is the corresponding physician’s estimated inverse productivity.

⁵¹The control sample is a nearly identical balanced panel of patients for randomly selected spells of other physicians with no reimbursement variation from certification (See Section 3.2).

⁵²Adding patient demographics does not increase R^2 and slightly increases the coefficient on predicted intensity.

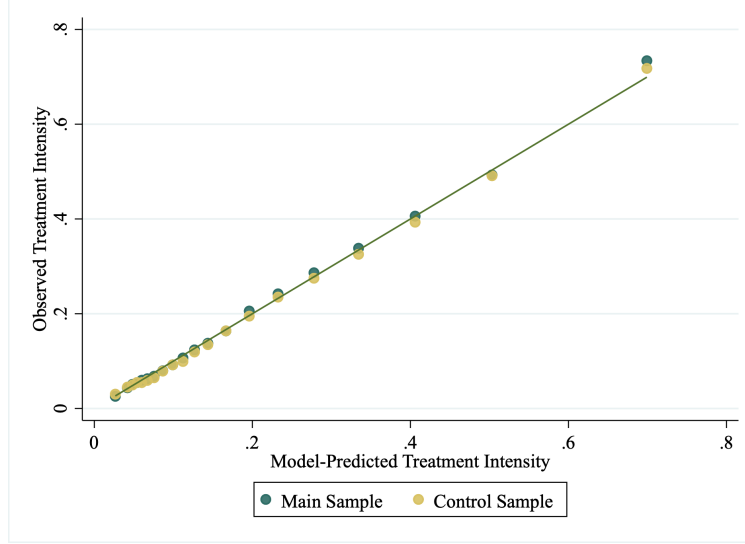
Table 3: Correlates of Physician Heterogeneity

	$\ln c$	$\ln \alpha$	$\ln \gamma$
Constant	0.902*** (0.168)	8.418*** (0.280)	-0.348*** (0.011)
Age	0.031 (0.028)	0.024 (0.048)	0.035*** (0.002)
Max Enrollment	-0.011 (0.032)	0.019 (0.052)	-0.015*** (0.002)
Pr(Diagnostic)	-0.057* (0.030)	0.023 (0.049)	-0.085*** (0.002)
Ever Fixed-Salary	0.113 (0.184)	-0.050 (0.297)	0.113*** (0.010)
Female	0.018 (0.060)	-0.049 (0.101)	-0.003 (0.004)
Migrant	-0.104* (0.063)	-0.022 (0.110)	-0.021*** (0.004)
Rural Municipality	0.099 (0.077)	-0.091 (0.127)	0.003 (0.004)
Trend	0.121 (0.304)	-0.639 (0.517)	-0.138*** (0.018)
S.D. Residual	0.227*** (0.031)	0.318*** (0.029)	0.145*** (0.002)
$\rho(\ln c, \ln \alpha)$	-0.269* (0.139)		
$\rho(\ln c, \ln \gamma)$	0.561*** (0.101)		
$\rho(\ln \alpha, \ln \gamma)$	-0.295** (0.137)		

Notes: This table regresses logs of physician-level estimates of cost c , altruism α , and inverse productivity γ on observable characteristics. Standard errors come from the delta method using the approximate Hessian of parameter estimates. Continuous demographics are normalized by mean and standard deviation relative to the full population of physicians. Max Enrollment is the largest number of patients a physician agrees to potentially have on their registered list. Pr(Diagnostic) is the share of reimbursement lines that are diagnostic relative to procedures. Ever Fixed-Salary indicates that physicians were at some point employees with a fee-for-service rate of zero. S.D. Residual is the standard deviation of the residuals from these regressions. ρ indicates the correlation between residuals.

*, **, and *** denote statistical significance at the 10%, 5%, and 1% level.

Figure 6: Model Fit: Ventiles of Predicted Treatment Intensity



Notes: This plot shows ventiles of predicted patient-month treatment intensity on the x-axis against means of actual treatment intensity on the y-axis. The 45-degree line is also plotted.

aligns with the covariance condition from Proposition 1. The upper panel of Table 3 shows that observed characteristics explain some of this variation.⁵³ For example, productive physicians tend to be younger and born outside of Norway. They hold larger lists of patients, make greater use of diagnostics relative to procedures, and historically worked under fee-for-service contracts. The bottom panel shows residual variation in physician types is also widely dispersed and correlated.⁵⁴

Observed patient characteristics explain a moderate share of the variation in treatment intensity by shifting illness severity (See Table A.3). Seasonality and particular chronic illnesses are major determinants of patients' treatment needs. For example, utilization is much lower in August than in January, and diabetes patients are more likely to visit a primary care physician than cancer patients. Other coefficients are precise but unexpectedly low in magnitude relative to raw correlations with treatment intensity, e.g., lagged treatment intensity and gender. Increasing lagged treatment by one standard deviation would only increase the health shock about as much as the average difference between January and April. This small coefficient reinforces the assumption that the distribution of health shocks is conditionally independent across months within each patient. Finally, conditioning on the full set of patient demographics, patient severity is highly dispersed and difficult to predict.

⁵³All standard errors are adjusted for noise in parameter estimates.

⁵⁴Rather than introduce a menu, a regulator could condition the reimbursement rate on observed physician characteristics. The substantial unobserved heterogeneity suggests that targeting observed characteristics may be ineffective. Likewise, targeting may be infeasible given, e.g., legal protections for age and physicians' collective bargaining.

6 Counterfactual Menus of Contracts

6.1 Baseline Counterfactuals

Using estimates, I simulate physicians’ choices under counterfactual menus to illustrate the welfare effects of self-selection. First, I quantify the cost of information asymmetry by solving for the personalized contracts offered by the regulator under perfect information. Second, I benchmark to the status quo and find that the existing reimbursement supplement is nearly optimal if the regulator can only offer a single (uniform) contract. Third, I demonstrate that even an arbitrary two-contract menu can increase welfare relative to a uniform contract because the distribution of physician heterogeneity satisfies key properties of dispersion and correlation. Fourth, I derive the menu of linear contracts that maximizes welfare given imperfect information. I conclude by assessing the equity implications of the optimal menu.

To scale health production into dollars, I assume that the regulator values incremental health production from certification as much as incremental expenditure. This assumption implies that the regulator is 3.1 times as altruistic as the median certified physician.⁵⁵ Table 4 compares aggregate health production across counterfactual menus, relative to the pre-certification status quo. Columns for expenditure and physician indirect utility reflect the budget and participation constraints. To focus on the role of reimbursement in treatment intensity, I fix other sources of variation at values six months before certification: enrollment, the share of patient types for each physician, and pre-certification fee-for-service rates. I provide additional detail on how I measure counterfactual outcomes and search for counterfactual menus in Appendix B.3.

With perfect information about physician heterogeneity, personalized contracts would increase expected health production by \$525 million per year nationally. In this first-best allocation, efficient contracts achieve nearly four times the gain in health production of the observed reimbursement rate increase while satisfying strict participation and budget constraints. I identify efficient contracts by selecting the fee-for-service rate for each physician from a grid that maximizes $E[\alpha_R h(m^*, \lambda) - p m^*]$. I set base payments so that in expectation, each physician is indifferent between the efficient contract and the status quo. Figure 7 shows substantial heterogeneity in the efficient reimbursement rates.⁵⁶ On average, efficient rates

⁵⁵For comparison, Gaynor, Mehta and Richards-Shubik (2023) calibrate a comparable parameter at 52.6 times the median altruism among providers based on the value of a statistical life-year. My approach does not internalize the regulator’s valuation of certification training beyond immediate changes to health production.

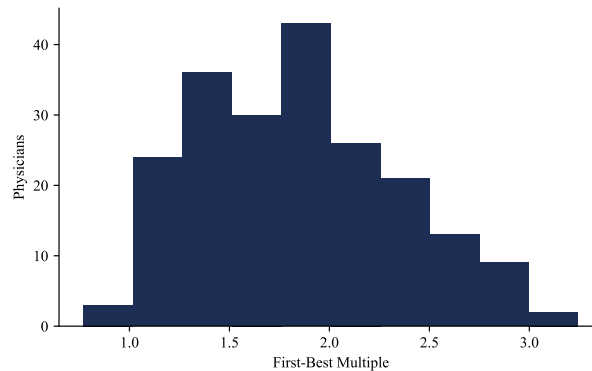
⁵⁶Throughout this section, I discuss each counterfactual fee-for-service rate as a multiple of the pre-certification rate. For example, 1.2 indicates 120 percent of the initial fee-for-service rate. This approach preserves variation in fee-for-service rates across patients while allowing simple graphical comparisons

Table 4: Annual Counterfactual Outcomes for Norwegian Population (\$M)

	Health Production	Share of Max	Expenditure	$E[V]$
Pre-Certification	0.0	0.000	0.0	0.0
Post-Certification	139.0 (0.4)	0.264 (0.001)	138.9 (0.4)	113.6 (0.4)
Efficient Contracts	525.8 (3.0)	1.000 (0.000)	137.2 (0.6)	0.0 (0.0)
Optimal Uniform Contract	153.7 (2.1)	0.292 (0.003)	132.5 (0.5)	103.6 (0.5)
Optimal Menu of Contracts	176.5 (1.9)	0.336 (0.003)	144.9 (0.4)	109.1 (0.6)

Notes: This table shows key outcomes from status quo and counterfactual contract menus, scaled annually to the Norwegian population (5.24M). All outcomes are based on ex-ante expectations over patient-months using the estimated distributions of physician types and patient severity, weighted across physicians by enrollment. Enrollment, the share of patient-types, pre-certification fee-for-service rates, and base payments are fixed at six months before certification. Post-certification fee-for-service rates are fixed at the month after certification. Counterfactuals vary the base payment and multiple of the status quo fee-for-service rate, enforcing participation and budget constraints. Health production is scaled such that the regulator is indifferent between incremental health production and incremental expenditure from certification. Share of Max divides the first column by its maximum from efficient contracts. Expenditure includes both fee-for-service and base payments. $E[V]$ is the expected indirect utility per patient-month of private physicians. Standard errors, shown in parentheses, are calculated across 25 bootstrap estimation samples, using randomly selection with replacement of patient-months by physician and re-solved counterfactuals. Figures A.9 and A.10 further illustrate the distribution of counterfactual contracts across bootstrap samples.

Figure 7: Dispersion in Efficient Reimbursement Rates



Notes: The y-axis is the count of physicians in each bin. The x-axis is a multiple of the pre-certification fee-for-service rate that maximizes health production subject to strict physician-level participation constraints and a global budget constraint (average expenditure must be less than status quo post-certification). Each base payment is the lowest level that satisfies each physician's participation constraint.

are 84 percent above the initial status quo rate with substantial variation ($SD = 49$ percent). Efficient rates are far above the status quo because a large share of physicians have high cost of effort or low altruism. In the status quo, these physicians spend relatively little time with patients, so the health benefits of incremental treatment are large.

In the status quo, the reimbursement rate increases by 24 percent which improves health production by approximately one-fourth as much as efficient rates. For a non-negligible share of physicians, the post-certification rate is higher than their efficient rates. For example, the most altruistic physicians do not change treatment intensity enough to justify the mechanical increase in expenditure. On the other hand, most physicians are far below their efficient rates, so the best uniform contract consists of an even higher fee-for-service rate and a lower base payment. The higher rate increases health production and the lower base payment decreases expenditure. The base payment can be lowered while satisfying the participation constraints because physicians have a high willingness to pay for the higher rate.

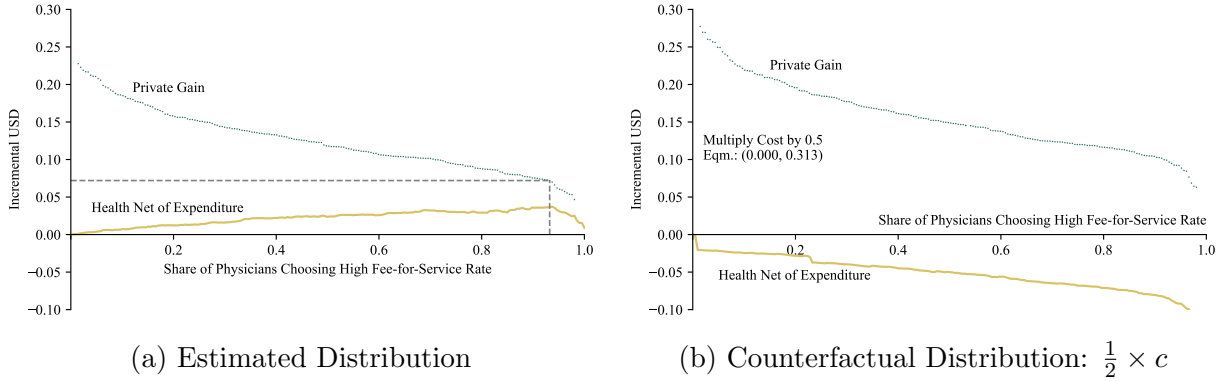
Even a two-contract menu achieves meaningful efficiency gains relative to the best uniform contract. Reinforcing the intuition from Section 2.2, this intermediate exercise shows how self-selection may not increase welfare for some distributions of physicians. I adapt the graphical framework for selection markets introduced in Einav, Finkelstein and Cullen (2010) and extended by Marone and Sabety (2022). I start with the optimal uniform contract (p_L) and add a contract (p_H) to the menu with a marginally higher reimbursement rate. If p_H requires accepting a relatively low base payment, then only a fraction of physicians with large willingness-to-pay ($WTP \equiv EV(p_H, 0) - EV(p_L, 0)$) will choose it. Physicians with large WTP have relatively low cost, high altruism, and high productivity (See Appendix C.1). However, these traits also predict relatively large increases in expenditure which might outweigh the corresponding increase in health production, especially if cost is low relative to altruism.

Figure 8a shows the tradeoff between increased health production and increased expenditure across physicians, ordering physicians in decreasing order by their WTP. The WTP curve is like a demand curve, indicating participation in the high-fee-for-service contract for various prices Δb , i.e., lower base payments. I also summarize welfare as incremental social surplus: expected health production minus expected expenditure, relative to the low-fee-for-service contract, where expenditure reflects both fee-for-service and base payment changes in equilibrium.⁵⁷ For each percentile of physicians choosing the high rate, I show the aver-

across counterfactuals. In a robustness check below, I consider a unique reimbursement rate (level) for each type of patient with similar observed characteristics.

⁵⁷At virtually any quantile of WTP, some physicians will be inefficiently selected into the high-fee-for-service contract and some will be inefficiently selected into the low-fee-for-service contract, relative to full information with the same restricted menu.

Figure 8: Two-Contract Menus: Setting Incremental Base Payments



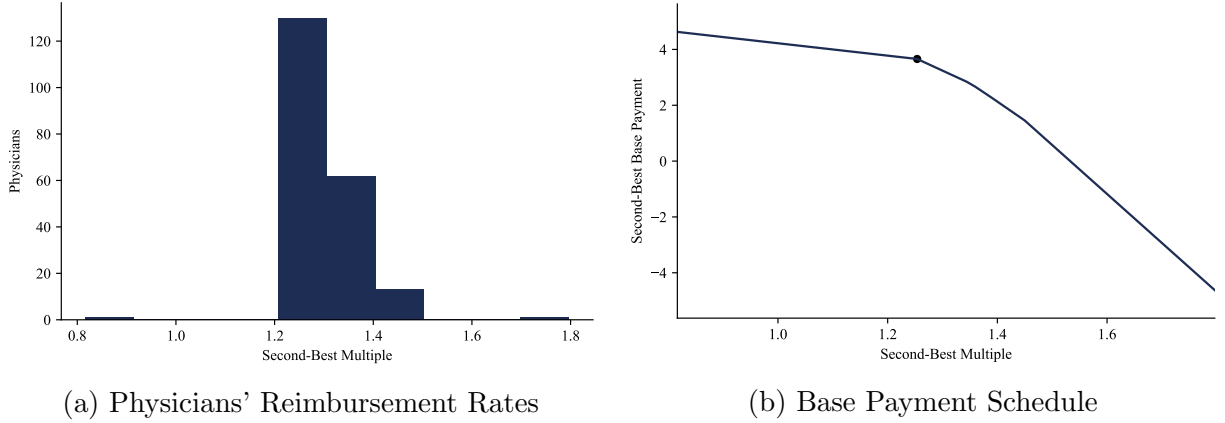
Notes: This figure shows outcomes under a menu that includes the best uniform fee-for-service rate and a fee-for-service rate that is incrementally higher while varying the difference in the base payment between these contracts. The x-axis orders a continuum of physicians according to their decreasing private gains from an increased reimbursement rate. The green line is incremental social surplus for each percentile of private gain: expected health production minus expenditure among all patients (and all physicians). Grey dashed lines indicate the optimal share of physicians choosing the high-fee-for-service contract and the corresponding difference in base payments between the two contracts. Panel A shows the estimated distribution of physician heterogeneity. Panel B multiplies estimated marginal cost by 0.5.

age incremental surplus across all patients. The regulator sets incremental base payments to maximize this quantity, where 93 percent of physicians choose the high-fee-for-service contract with a \$0.07 lower base payment. With smaller differences in the base payment, the 7 percent of physicians with lowest WTP would increase expenditure more than health production. Figure A.8 shows that variation in social surplus is best explained by cost, and variation in WTP is best explained by mean patient severity.

Figure 8b illustrates a counterexample where the two-contract menu is not more efficient than the uniform contract. This panel repeats the previous exercise with a counterfactual distribution of physician heterogeneity. When each physician's cost of effort is half as large, WTP is greater and efficient rates are lower. WTP and social surplus are not sufficiently correlated for both contracts to be traded. Incremental expenditure would always exceed incremental health production. The regulator sets the incremental base payment high enough for a corner solution where all physicians choose the low rate. Lower cost of effort makes efficient rates relatively similar to the best uniform contract, i.e., small allocative effects and large inframarginal rents.

The optimal 7-contract menu achieves large efficiency gains by separating some physicians into high-rate contracts, equivalent to \$32 million per year more than the status quo. To search for this menu, I sequentially adjust each potential contract's base payment to maximize equilibrium social surplus until convergence (see Appendix B.3). Most physicians choose just one of three contracts (Figure 9a) and the optimal base payment decreases con-

Figure 9: Optimal Menu of Contracts



Notes: In Panel A, the y-axis is the count of physicians in each bin. The x-axis is a multiple of pre-certification fee-for-service that maximizes health production subject to strict physician-level participation constraints and a global budget constraint (average expenditure must be less than status quo post-certification). Panel B plots base payments versus multiples of status-quo fee-for-service rates for the optimal menu. The point in Panel B indicates the optimal uniform contract.

cavely in the fee-for-service rate as expected (Figure 9b).⁵⁸ Perhaps a smaller menu would involve lower implementation costs: Figure A.7 shows that while increasing the number of contracts per menu generally improves welfare, most efficiency gains can be achieved with a small number of contracts.

Redistribution across patients drives some of the gains in average welfare from efficient contracts and the optimal menu. To explore redistribution, I disaggregate counterfactual outcomes across physician types. In Table 5 I categorize physicians into 16 groups based on whether each of cost c , altruism α , inverse productivity γ , and expected patient severity $\lambda \sim F$ are above or below the median. The largest health production increases are among the 36 percent of physicians with large allocative effects from cost and altruism, combined with large inframarginal rents (and WTP) from effective severity. Changes in expenditure are relatively small. All else equal, these physicians tend to have low private gain from increased rates. The menu can separate some of these physicians into high fee-for-service rates because productivity and average patient severity are highly dispersed.

6.2 Robustness

Relaxing restrictions on model assumptions and sample construction suggests that the efficiency of self-selection does not rely on an idiosyncrasy of the empirical approach or setting.

⁵⁸Moreover, Figure A.10 shows that across bootstrap samples, the optimal menu consistently lies on approximately the same curve of Base Payment versus Fee-For-Service Multiple.

Table 5: Counterfactual Outcomes by Physician Type

Physicians		Efficient Contracts		Menu of Contracts		
Type	Share	$\Delta E[h(m)]$	$\Delta E[p m + b]$	$\Delta E[h(m)]$	$\Delta E[p m + b]$	$\Delta E[V(p)]$
$c_L, \alpha_H, \gamma_L, F_L$	0.171	1.311	0.554	0.855	1.505	1.177
$c_H, \alpha_L, \gamma_H, F_L$	0.160	14.165	3.420	2.984	1.545	1.161
$c_H, \alpha_L, \gamma_H, F_H$	0.155	20.003	4.486	5.793	3.140	2.303
$c_L, \alpha_H, \gamma_L, F_H$	0.154	1.441	0.622	1.155	2.549	2.012
$c_L, \alpha_L, \gamma_H, F_L$	0.049	3.860	1.367	1.749	1.963	1.527
$c_H, \alpha_L, \gamma_L, F_H$	0.047	21.554	4.820	4.905	2.010	1.412
$c_L, \alpha_H, \gamma_H, F_H$	0.045	3.201	1.251	2.336	4.003	3.155
$c_H, \alpha_H, \gamma_L, F_L$	0.037	5.670	1.977	1.722	1.567	1.206
$c_L, \alpha_H, \gamma_H, F_L$	0.033	2.172	0.847	1.218	2.031	1.630
$c_H, \alpha_H, \gamma_L, F_H$	0.031	6.620	2.372	2.468	2.529	1.953
$c_H, \alpha_H, \gamma_H, F_L$	0.025	5.997	2.052	1.942	2.064	1.654
$c_H, \alpha_L, \gamma_L, F_L$	0.022	7.306	2.315	2.157	1.366	0.977
$c_H, \alpha_H, \gamma_H, F_H$	0.019	6.240	2.288	3.405	3.974	3.067
$c_L, \alpha_L, \gamma_L, F_L$	0.017	5.755	1.285	6.489	2.202	1.129
$c_L, \alpha_L, \gamma_L, F_H$	0.017	6.694	1.690	10.557	4.990	2.875
$c_L, \alpha_L, \gamma_H, F_H$	0.017	4.359	1.639	3.106	3.880	2.872

Notes: This table shows average outcomes for efficient contracts and for the optimal menu, disaggregated across groups of physicians (rows). For physician types, the subscript "H" indicates above-median, and "L" indicates below median. Physician type is a combination of physicians' cost c , altruism α , inverse productivity γ , and expected patient severity $F(\lambda)$. $\Delta E[h(m)]$ represents the change in health production relative to the status quo. Likewise, $\Delta E[p m + b]$ represents incremental expected expenditure and ΔEV represents incremental expected indirect utility. Outcomes are averages across patient-months within each group, measured in USD.

First, I find evidence for external validity within Norway: including out-of-sample physicians in counterfactuals does not change the main finding. The model predicts that physicians would select into the certification sample based on observed and unobserved characteristics. Table 1 shows that non-certified physicians have slightly higher treatment intensity which may be explained by relatively old and chronically ill patients. Non-certified physicians are also more likely to be old, born outside of Norway, and use diagnostics. To explore unobserved differences for non-certified physicians, I estimate the distribution of unobserved heterogeneity based on the relatively weak assumption that non-certified physicians have the average log altruism among certified physicians with identical observed characteristics. This assumption is necessary because the identification of altruism requires observing the same physician with different fee-for-service rates. Physicians in the main estimation sample can still be selected on observed heterogeneity in altruism and both observed and unobserved

heterogeneity in cost and productivity. Reinforcing this assumption, unobserved heterogeneity in altruism is precise and small relative to the mean (Table 3). Likewise, Table A.4 shows that estimates fit observed treatment intensity well for both samples. If non-certified physicians were meaningfully selected on unobserved heterogeneity in altruism, predicted treatment intensity would be a weak predictor of observed treatment intensity. Finally, repeating the counterfactual analysis for the combined population of certified and non-certified physicians results in similar outcomes.⁵⁹

Second, I find evidence for external validity outside of Norway: even large perturbations of estimates rarely change the main finding. In Table A.5, I first perturb cost c , altruism α , and productivity γ^{-1} . Generally, a menu increases efficiency when scaling estimates, removing one or two dimensions of unobserved heterogeneity (by replacing estimates with the sample mean), or limiting dispersion in estimates by halving variance or dropping outliers. Consistent with Proposition 1, when doubling the variance of cost or altruism, I cannot find efficiency gains from a menu. The variance of severity σ_λ and regulator altruism α_R have large impacts on the levels of counterfactual outcomes. With sufficiently low regulator altruism, i.e., high inframarginal rents and low efficient rates, a uniform contract is optimal.

Robustness to scaling variance and dropping outliers of physician heterogeneity also suggests that estimation error does not drive the main findings. With overestimated heterogeneity in cost, altruism, and productivity, the gains from self-selection might appear artificially large. Likewise, bootstrapped standard errors are small across aggregate counterfactuals outcomes (Table 4), physician-specific contracts (Figure A.9), and the relationship between the reimbursement rate and base payment in the optimal menu (Figure A.10).

Third, descriptive evidence reinforces the exclusion assumption that high-severity patients do not systematically choose particular physicians. The assumption simplifies the analysis by avoiding dynamic considerations. In practice, patients can freely switch between physicians whose enrollment is below the contracted maximum, up to twice per year. As a result, physicians might perceive a link between current treatment intensity decisions and future enrollment, e.g., through reputation effects, which would increase future revenue. Likewise, patients with higher unobserved severity might systematically sort towards physicians with higher expected health production (low cost, high altruism, high productivity). Descriptive evidence suggests that these are not first-order concerns. Figure A.1 shows that enrollment and the share of enrolled patients that are over 60 or chronically ill do not systematically vary with certification, unlike treatment intensity and health produc-

⁵⁹See Table A.5. Specifications that pool different estimation samples should be interpreted with caution. Results are also similar when excluding rural physicians, who anecdotally sometimes face unusual circumstances.

tion.⁶⁰ Enrollment and the share of patients with higher treatment need should increase if patients are sorting towards physicians with greater health production after certification due to increased treatment intensity. Likewise, as shown in Section 3.3, physicians’ fixed effects in treatment intensity are highly dispersed whether estimated among all patients or only quasi-randomly assigned patients. To test for medium-run sorting, I regress an indicator for switching physicians in the next six months on model-predicted health production, patient demographics, and fixed effects for year and calendar month.⁶¹ Column (2) of Table A.6 shows that the correlation between health and switching to a new physician is imprecise, with point estimates that are small in magnitude. By contrast, expected health production is predictive of (lower) future avoidable hospitalizations and mortality. In Figure A.11, raw means also suggest a decline in avoidable hospitalization after three years. Likewise, cumulative mortality is 36 percent lower than among patients of non-certified physicians.

Fourth, motivated by Ellis and McGuire (1986) and McGuire and Pauly (1991), I test for income effects – nonlinear cost of effort – with a likelihood ratio and cannot reject the baseline model. Income effects can also rationalize why some physicians reduce treatment intensity by a small amount in response to newly registered patients (Barash, 2024) or an increase in reimbursement rates (Figure 3). To estimate physicians’ marginal disutility of expected workload, I extend the theoretical framework and estimation strategy with additional assumptions, detailed in Appendix A.1. If income effects do exist, they seem too small relative to unobserved variation in patient severity to be economically meaningful. Figure A.13 tests the related assumption that physicians do not face binding capacity constraints. Over ten years, the distribution of physicians’ monthly treatment intensity varies smoothly near each physician’s maximum. Monthly treatment intensity should bunch at high values if some physicians occasionally reach capacity, e.g., due to idiosyncratic variation over time in the number of patients or realized severity. Next, Figure A.12 shows that the treatment intensity of high-altruism and low-altruism physicians is similarly responsive to the shock of a first avoidable hospitalization. This suggests that estimates of high altruism are not biased by an unobserved constraint. Likewise, the across-time variance of pre-certification workload is similar for low- and high-altruism physicians.⁶² I do not find evidence that patients of

⁶⁰Figure A.2 shows that certified and non-certified physicians experience similar trends in enrollment for at least two years after certification.

⁶¹I use model estimates to calculate expected health production for each patient in the main estimation sample during the six months post-certification. I measure switching 7-12 months after certification.

⁶²I aggregate hours for each physician in each month before certification and then calculate the across-month variance. This physician-specific variance does not correlate precisely with estimated altruism. If some physicians are less responsive to certification because of capacity, then low altruism should correlate with low variance. Such physicians would work a similar amount each month (at capacity).

high-altruism physicians are more likely to seek treatment elsewhere.⁶³ Finally, as shown in Table A.5, the optimal menu of contracts leads to similar welfare gains over a uniform contract when I impose a capacity constraint and repeat counterfactuals.⁶⁴ The constraint limits large expenditure increases on high-severity patients when the health production curve is relatively flat, while high rates still permit large gains for less severe patients. As a caveat, physicians might respond to counterfactual contracts by spending less time on other work, e.g., at nursing homes or universities, that is socially valuable but unmeasured.⁶⁵

Finally, counterfactual outcomes are nearly identical with an alternate parameterization of health production from the insurance literature (Cardon and Hendel, 2001; Einav et al., 2013; Marone and Sabety, 2022). Those papers use a quadratic function with a linear term which results in a convenient expression for second-stage choices: $h_0 + h_1(m - \gamma\lambda) - \frac{h_2}{2}(m - \gamma\lambda)^2$. In the baseline approach, I assume $h_1 = 0$ because it is not separately identified from the mean of private marginal cost apart from functional form.⁶⁶ To test the alternate parameterization, I re-estimate the model with $h_1 \geq 0$. I focus on non-negative values because previous studies estimate a parameter close to 1, and health production should initially increase in treatment. I estimate $h_1 = 0.073$.

6.3 Extensions

When considering more flexible contract structures, I find mixed evidence of benefits relative to the menu of linear contracts. Appendix A.2 and A.3 include a more detailed discussion. First, offering multiple menus produces similar outcomes to the single baseline menu. I allow the regulator to target menus to chronically ill patients then to rural patients since each group experienced large health improvements from the baseline menu. As an aside, that baseline heterogeneity highlights an underexplored dimension of rural health disparities: in the status quo, the most rural patients are more than twice as far from the full-information benchmark as the most urban patients. Second, a nonlinear uniform contract further increases welfare only when most physicians are allowed to be made worse off. I extend the method in Gaynor, Mehta and Richards-Shubik (2023) to derive a contract where the marginal reimbursement rate varies with how much treatment has already been provided to a patient.

⁶³Patients registered with high-altruism physicians receive relatively little primary care from secondary opinions and urgent care centers. If the registered physician was capacity-constrained, patients might seek more treatment from other physicians.

⁶⁴In this case, I bound workload (total simulated hours per physician-month) below the 99th percentile reached after certification.

⁶⁵Table A.5 shows similar gains to a menu when excluding physicians that initially work part-time, i.e., those who spend fewer than 25 hours per week with registered patients.

⁶⁶ h_0 is also not identified but does not affect choices. h_2 is absorbed in altruism.

The contract is approximately linear beyond low levels of treatment intensity, redistributing away from patients with very low severity to most other patients. Third, a regulator might further improve patient health through policies that complement contracts by shifting the allocation of patients across physicians. In a stylized example with two physicians, jointly determining rates and patient switches can increase incremental social surplus by 17 percent relative to efficient reimbursement rates alone. Policies that change the long-run distribution of physician traits could also be complementary, like public subsidies for support staff or telehealth to lower cost of effort, performance benchmarks to increase altruism, and promotion of long-term patient-physician relationships to increase productivity via soft knowledge.

7 Conclusion

In this paper I present a framework for deriving the optimal menu of physician reimbursement contracts. The framework incorporates unobserved patient illness severity and physicians' endogenous choices of contract and treatment intensity. I characterize the conditions on multidimensional physician heterogeneity under which self-selection among a menu of contracts is more efficient than a uniform reimbursement contract. These conditions are met in the empirical example of Norwegian primary care physicians. I estimate the distributions of physician and patient heterogeneity, exploiting the sudden large variation in the fee-for-service rate when physicians become certified as general practitioners. I find large efficiency gains from introducing self-selection, and that finding is robust to several model enrichments, estimate perturbations, and alternative samples.

The most direct policy implication is that the Norwegian National Insurance Scheme could cost-effectively improve access to primary care by offering a menu of 2-7 linear contracts. These contracts are easy to understand because they have the same linear structure as status quo reimbursement. The difference is that each contract exchanges a higher multiple on service-level reimbursement for lower revenue per registered patient-month. The regulator could use its existing data and infrastructure to administrate the policy counterfactual as an occasional settlement payment. Moreover, the menu of contracts is efficient even as a voluntary reform: physicians can still choose the status quo contract, which might make it acceptable to the association that negotiates reimbursement on behalf of physicians. I also find descriptive evidence of reductions in hospitalization and mortality. By contrast, economic theory and empirical evidence alike predict that Norway's initiative to increase base payments for relatively ill patients will not immediately affect treatment intensity, because marginal incentives are unchanged.⁶⁷

⁶⁷On the other hand, such a reform may effectively deter exit in the long term. With sufficient exit, capacity

Beyond Norway, my framework for evaluating the efficiency of self-selection is broadly applicable to settings featuring heterogeneous altruistic agents that experience panel variation in marginal reimbursement. In healthcare, this includes systems in which many physicians derive most revenue from contracts with a single payer, e.g., several countries’ health agencies or Kaiser Permanente in the United States. External validity might be limited in settings where prices are negotiated or patients frequently switch physicians based on reputations for treatment intensity. Outside of healthcare, menu design may be an effective tool in the markets for indigent defense attorneys, K-12 educators, and social workers. These agents are likely altruistic – sacrificing some profit to improve outcomes for their clients and students – and also heterogeneous in marginal cost and productivity. The frequent lack of compensation for incremental effort may also contribute to capacity constraints and disparities in outcomes. My framework uses reimbursement variation, which often exists in these settings, but even cross-sectional data can be sufficient with additional assumptions.⁶⁸

Why are uniform contracts ubiquitous if the potential gains from self-selection are large? First, variation in incentives across physicians may conflict with norms concerning uniformity. Moreover, before considering multidimensional unobserved physician heterogeneity, policy-makers may not find it intuitive that increasing dispersion in patients’ treatment could be efficient. Second, there may be fixed costs of introducing counterfactual menus, e.g., costly experiments in reimbursement variation to derive the optimal menu or incremental costs of negotiation with a physicians’ union.

I also explore related applications of the model that may be productive directions for future research. Consistent with existing evidence that patients imperfectly perceive physician quality, I find evidence that patients may not be optimally allocated across physicians to maximize cost-effective health production. Future work might consider combining ongoing reimbursement contracts with policies that steer where physicians establish or where patients register. I also do not find evidence of income effects or capacity constraints in Norway, but these features may add nuance to contracting in related settings. Finally, the model can be used to decompose variation in healthcare utilization, complementing the literature on place-based effects and physician workforce policies.

constraints may bind and reduce treatment intensity.

⁶⁸For example, a simulation-based estimator could recover a parametric distribution of altruism with cross-sectional variation in reimbursement under a stronger exclusion assumption. Client severity must be conditionally independent of agent type and reimbursement, which is unlikely if, e.g., high-quality agents receive higher reimbursement. See Biasi (2021) or Hanushek et al. (2023) for reimbursement variation among teachers.

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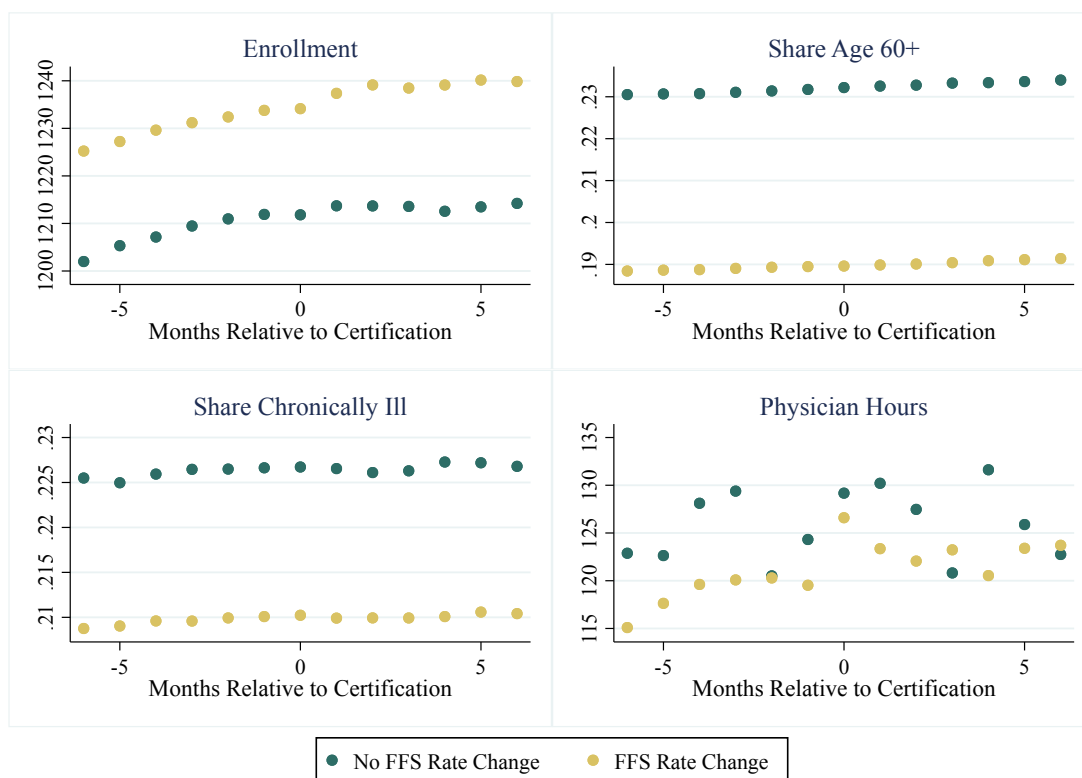
Should Physicians Choose Their Reimbursement Rate? Menu Design for Physician Payment Contracts

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Supplemental Appendix

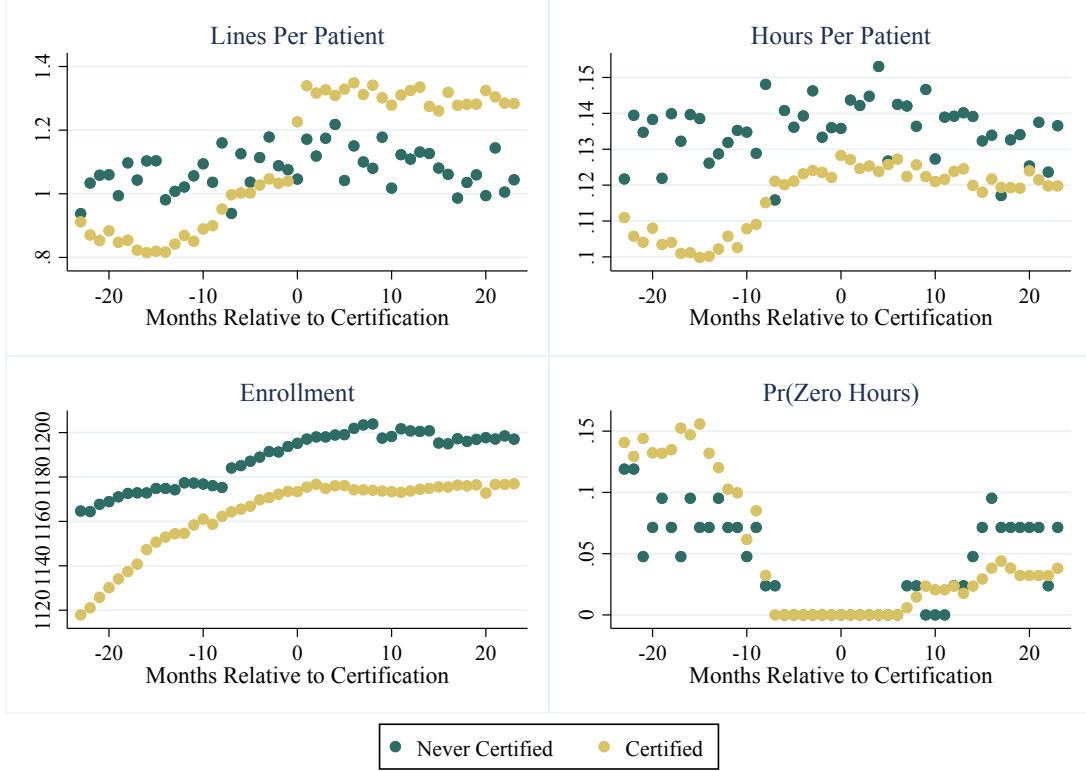
A Additional Analysis

Figure A.1: Raw Means of Characteristics Relative to Certification



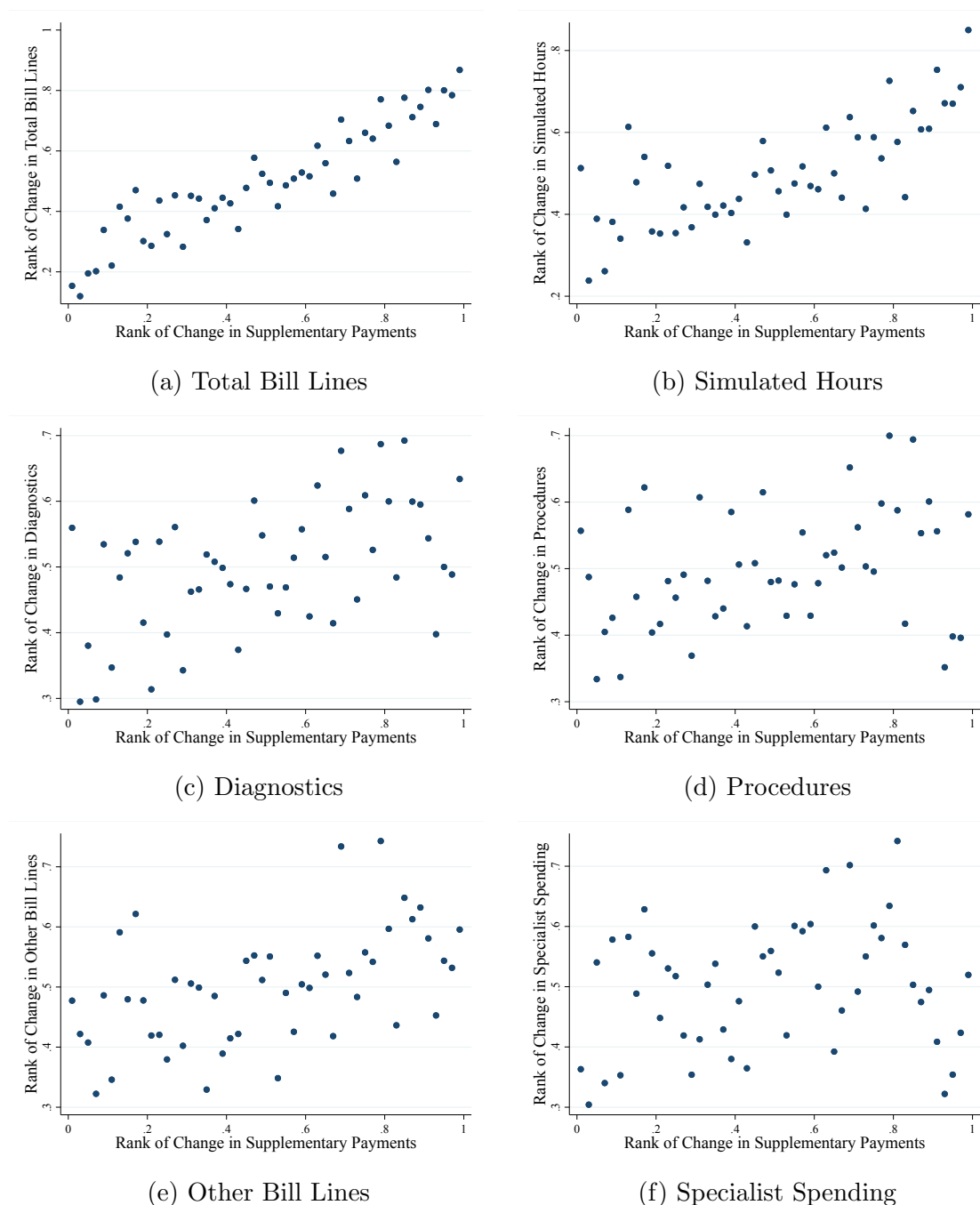
Notes: These plots show averages of treatment intensity outcomes across patient-months in the estimation and control samples in each month relative to certification. Each sample is a balanced panel of patients, and in the estimation sample, Month 0 is the first month in which the registered physician received a certification supplement.

Figure A.2: Long-Run Means Relative to Certification



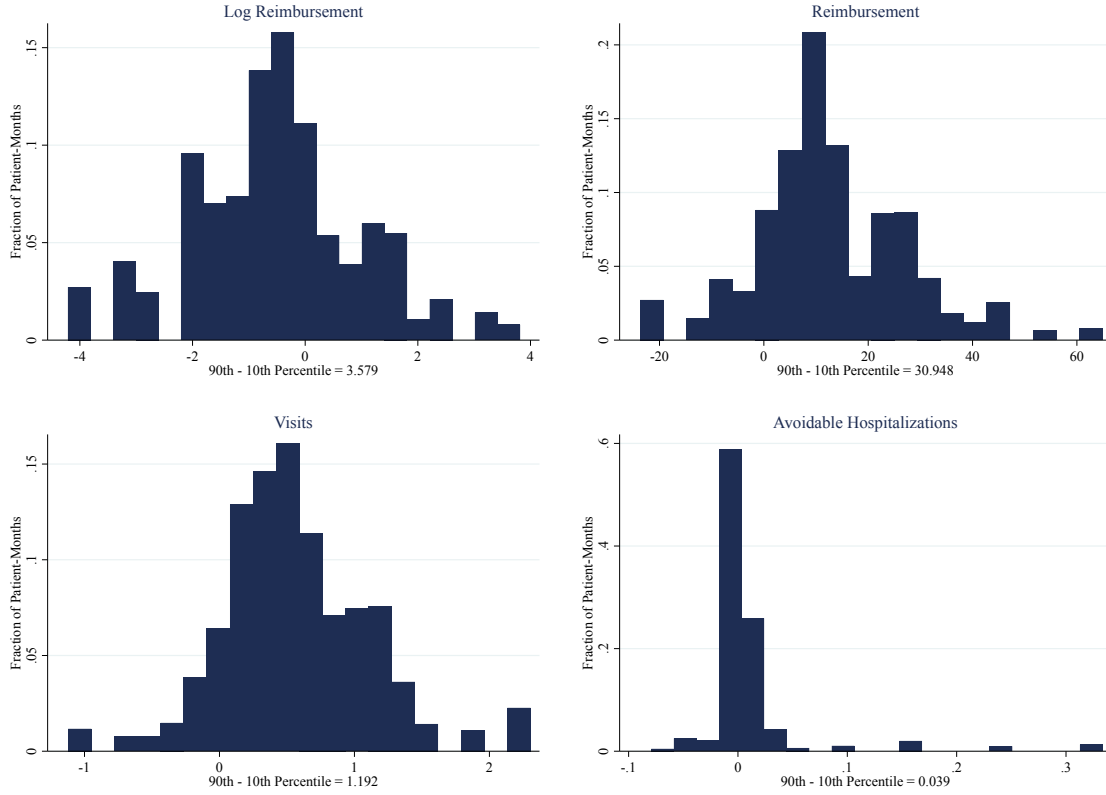
Notes: These plots show averages across physicians in the estimation and control samples in each month relative to certification. Each sample is a panel of physicians, and in the estimation sample, Month 0 is the first month in which the registered physician received a certification supplement. Unlike in other analyses like Figures 1 and A.1, each observation used to generate plots reflects aggregate labor supply of physicians, rather than the subset of treatment in a balanced sample of registered patients. Physicians must be registered to at least one list in each of the 49 months, but that list may change and labor supply may be zero in a given month. Bill lines include the certification supplement. Per Patient indicates that the aggregate is divided by the total number of registered patients.

Figure A.3: Similarity of Physician-Specific Responses to Certification Across Treatment Types



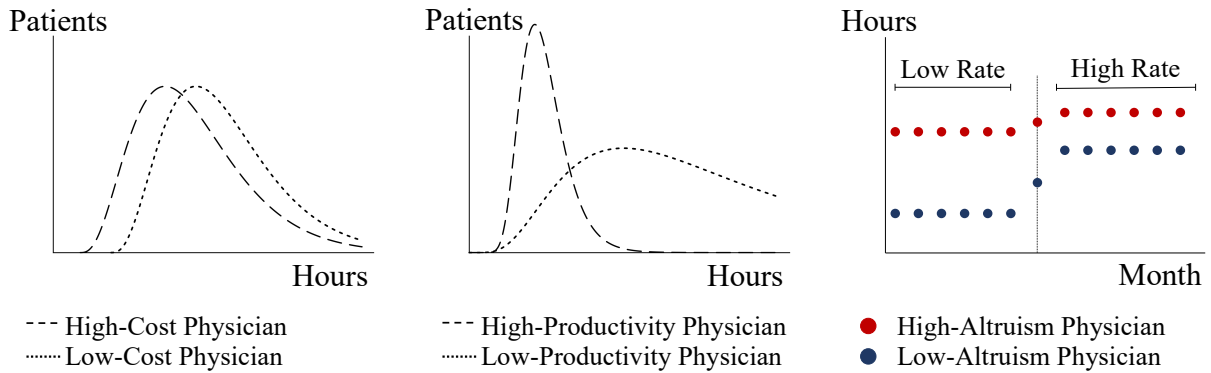
Notes: These plots show that physicians with relatively large increases in supplementary payments post-certification also have relatively large post-certification increases in other measures of treatment intensity. I first take means across patient-months in the six months pre-certification and the six months post-certification, for each physician. Next, I calculate the percentile rank across physicians of $Post - Pre$. Each panel use a different treatment measure to construct the y-axis. Each point is a mean for one of 50 quantiles along the x-axis. The sample includes certified physicians.

Figure A.4: Shrunk Assignment Effects for Certified Physicians



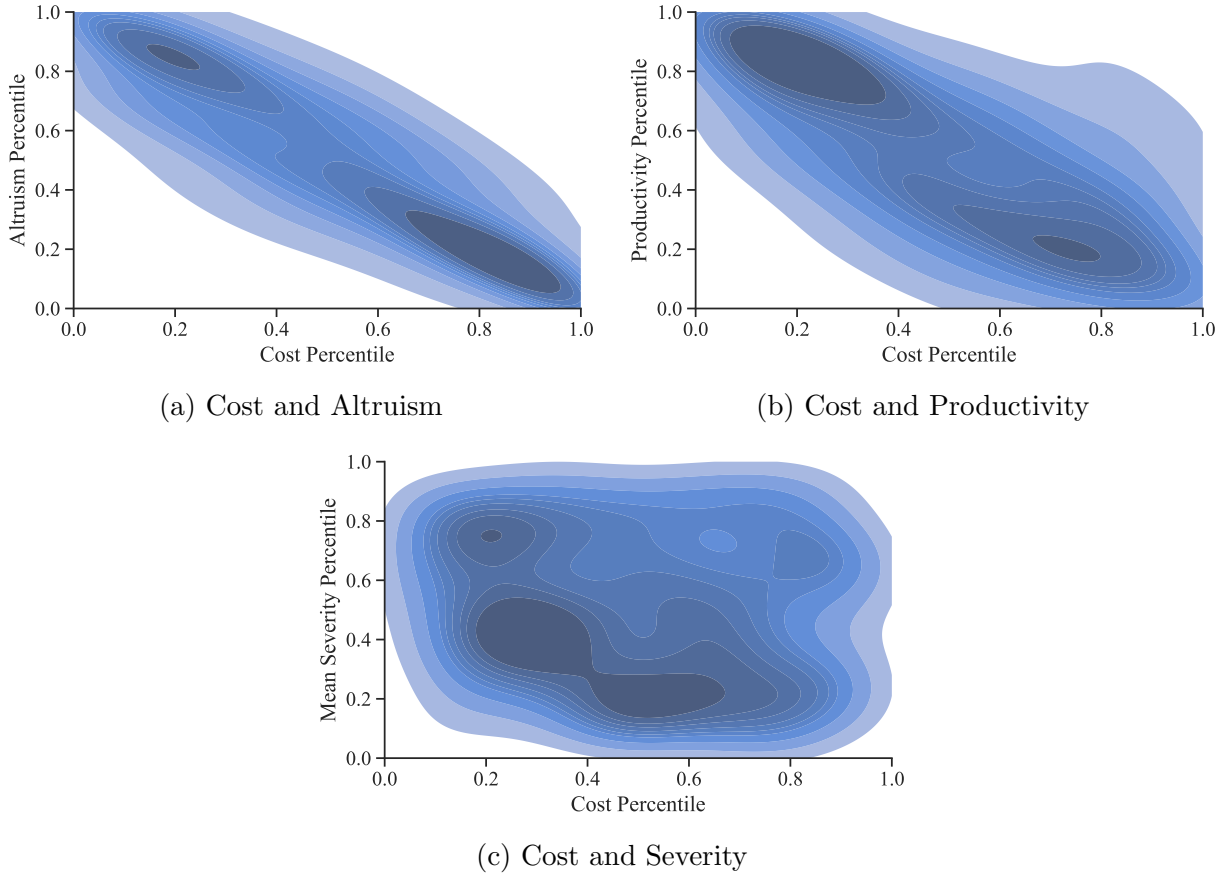
Notes: These histograms show the distribution assignment effects among physicians in the main estimation sample. Following Jinja et al. (2025), I estimate assignment effects by comparing patients from the same exiting physician who are conditionally randomly assigned to various focal physicians. Assignment effects are focal physician fixed effects from a regression including fixed effects for the exiting physician and calendar year. To reflect conditional randomness, I add controls for focal physician availability and an indicator for the same municipality. All estimates are shrunk to the mean using Empirical Bayes. All dependent variables are measured per patient-month during the six months after assignment.

Figure A.5: Stylized Example of Identification Intuition



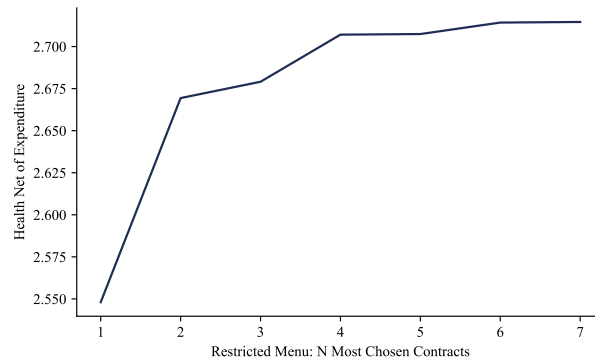
Notes: These plots illustrate the identification intuition of physician heterogeneity for the main specification ($\sigma = 0$). All else equal, cost represents a level shift in the distribution of treatment intensity, productivity increases the dispersion of that distribution, and altruism lowers responsiveness to fee-for-service rates.

Figure A.6: Distribution of Physician Heterogeneity



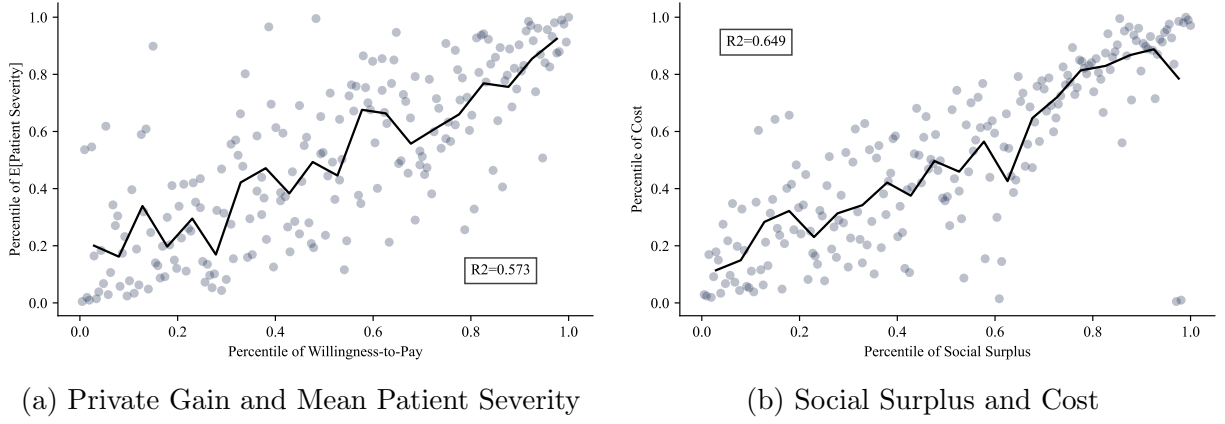
Notes: These plots summarize the joint distribution of estimated cost, altruism, and productivity across the estimation sample. Darker regions indicate higher density.

Figure A.7: Restricted Menus Achieve Less Welfare



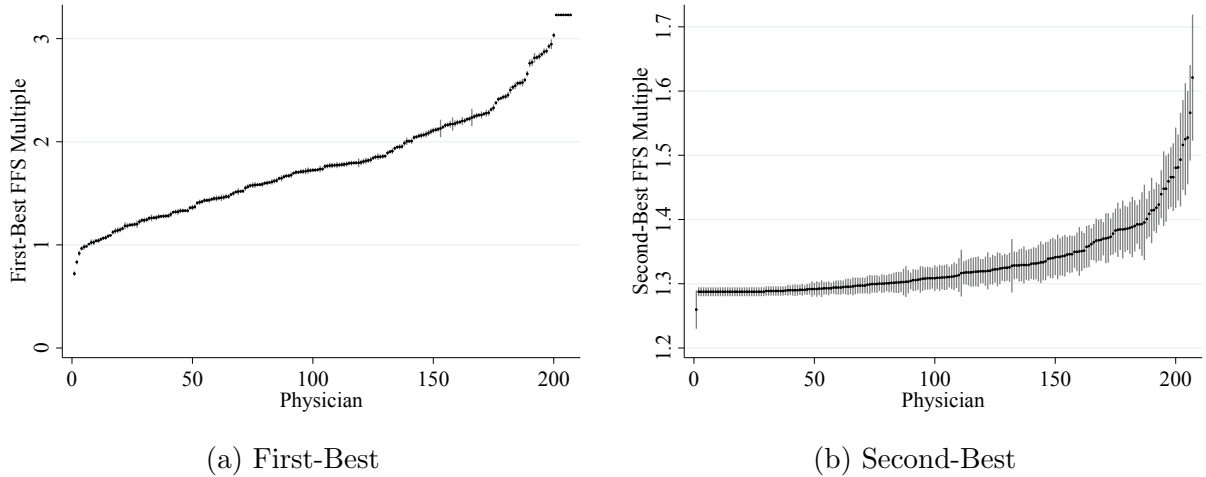
Notes: The y-axis is expected health production net of expenditure. The x-axis is the number of contracts per menu. For each menu, I re-solve for optimal base payments. I focus the search on the optimal menu's N most chosen contracts. I restrict this function to be non-decreasing when setting the base payment for the marginal contract.

Figure A.8: Two-Contract Menus: Correlations with Physician Type



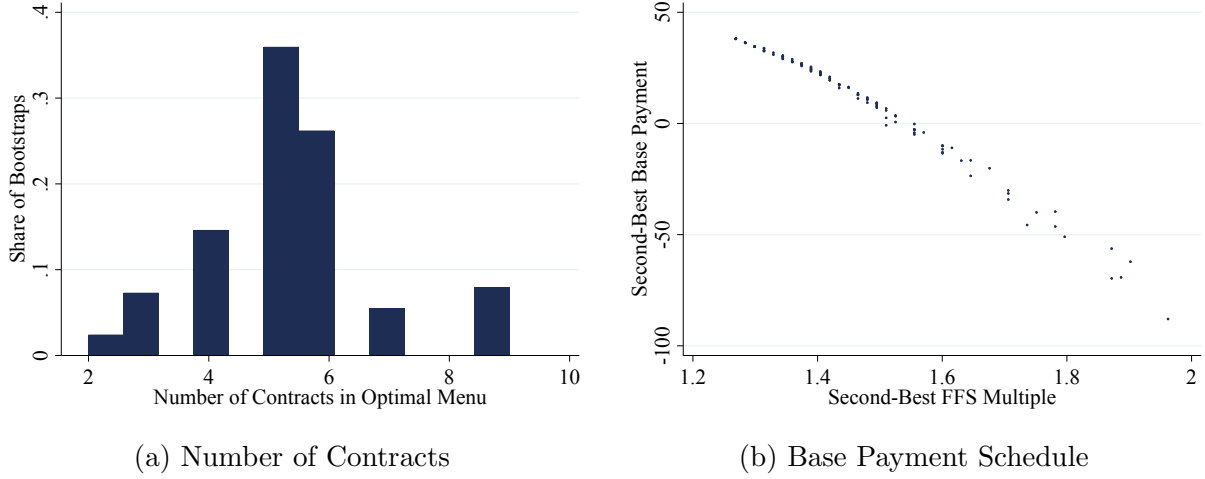
Notes: This figure plots, across physicians, the correlation between each incremental outcome from the two-contract menu in Figure 8a and its strongest predictor by bivariate R^2 . I separately regress the outcomes (private gain and social surplus) on percentiles of each dimension (cost, altruism, production). The R^2 statistics for private gain are 0.038 for c , 0.010 for α , and 0.041 for γ . The R^2 statistics for social surplus are 0.588 for α , 0.097 for λ , and 0.096 for mean patient severity. Private gain is the difference in expected indirect utility between the high- and low-fee-for-service contracts. Social surplus is the difference between contracts in expected health production minus expenditure.

Figure A.9: Physician-Specific Contracts Across Bootstrap Samples



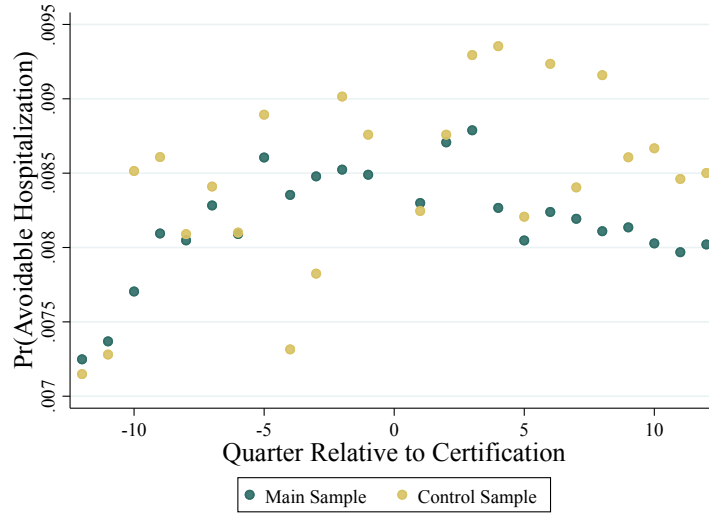
Notes: This figure plots the distribution of fee-for-service multiples across bootstrap samples for each physician. The x-axis is sorted separately for each panel, by mean Fee-For-Service Multiple. Error bars represent the bootstrapped 95 percent confidence interval. In each bootstrap sample, patient-months are randomly selected with replacement to maintain the original sample size per physician. In First-Best, the regulator has perfect information about physician types and offers each physician her efficient rate. In Second-Best, given imperfect information, the regulator designs the optimal menu of contracts and each physician self-selects a contract.

Figure A.10: Optimal Menu Across Bootstrap Samples



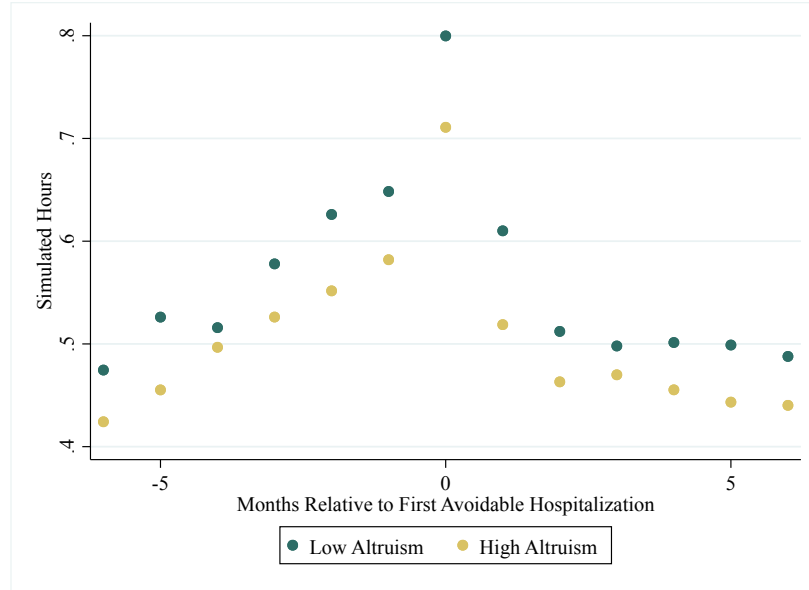
Notes: The left panel plots the distribution of the number of contracts in the optimal menu across bootstrap samples. Only contracts selected by two or more physicians are included. The right panel plots every optimal menu contract selected by two or more physicians, pooled across bootstrap samples. A contract is a pair of the reimbursement rate multiple and base payment. In each bootstrap sample, patient-months are randomly selected with replacement to maintain the original sample size per physician. In Second-Best, given imperfect information, the regulator designs the optimal menu of contracts and each physician self-selects a contract.

Figure A.11: Certification and Avoidable Hospitalization



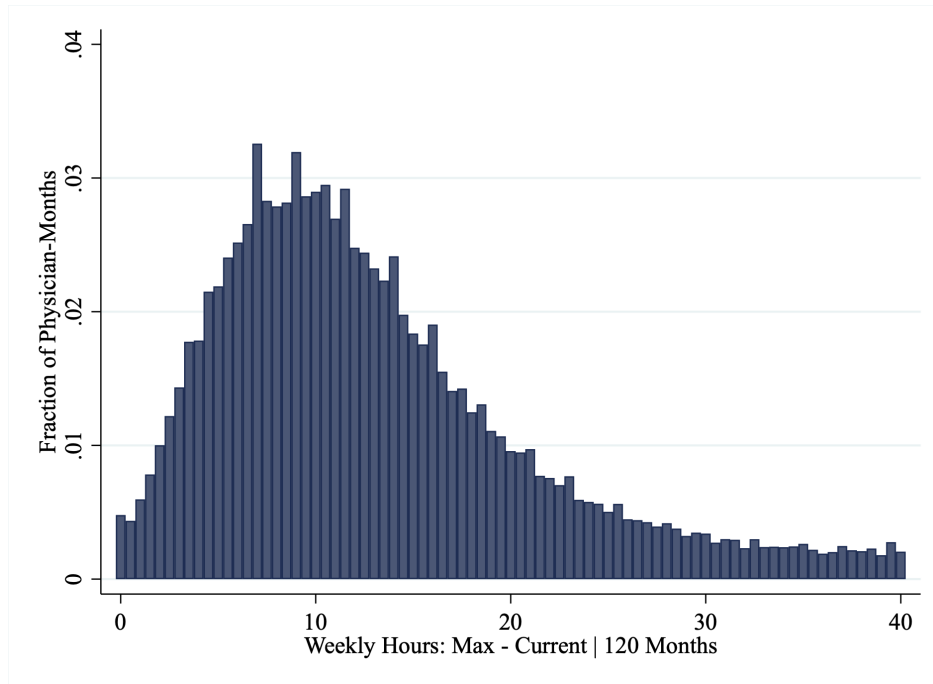
Notes: The plot shows the share of patients with an avoidable hospitalization in each quarter over six years. Each point is a mean across physicians. The underlying data includes patients who were consistently registered in the six months surrounding certification. The denominator of the physician-quarter patient share is the number of patients who are alive. Outside of Quarters -2 to 2, patients are not necessarily registered to the focal physician. The Main Sample includes certified physicians and the Control Sample includes non-certified physicians with randomly selected focal months. Both samples are restricted to physicians with certification dates in 2011-2014 so that utilization data exists in across all 72 months.

Figure A.12: Treatment Intensity Responds to Health Shocks



Notes: This figure shows average simulated hours across patient months in the six months before and after each patient's first avoidable hospitalization. The sample includes pre-certification patient-months for a balanced panel of consistently registered patients and is subset by whether the registered physician's estimated altruism is above or below the sample median.

Figure A.13: Capacity Constraints: Hours Do Not Bunch Near Each Physician's Maximum



Notes: This figure shows the distribution of transformed hours per week ($\tilde{M}_{jt}$) across physician-months ($j - t$). The transformation is $\max_t M_{jt} - M_{jt}$. The x-axis is truncated at 40 and I exclude the first month when a physician works the maximum number of hours.

Table A.1: Registered Patient Summary Statistics versus Population

	Population	Estimation Sample					
	Mean	Mean	Std. Dev.	% > 0	10th	50th	90th
Patient Characteristics							
Age	38.436	37.225	22.684	1.000	6.417	36.250	68.417
Female	0.495	0.505	0.500	0.505	0.000	1.000	1.000
Chronic Illness	0.200	0.210	0.407	0.210	0.000	0.000	1.000
Disability	0.060	0.064	0.244	0.064	0.000	0.000	0.000
Physician Characteristics							
Enrollment	1297.232	1235.749	314.715	1.000	880.000	1197.000	1592.000
Female Physician	0.356	0.438	0.496	0.438	0.000	0.000	1.000
Physician Age	49.000	39.777	6.123	1.000	33.500	38.083	49.500
Migrant Physician	0.215	0.226	0.418	0.226	0.000	0.000	1.000
Patients	5525876	215529					
Physicians	4769	207					

Notes: Summary statistics reflect patients' monthly totals. The Population column reflects all Norwegian patients in 2012. All other columns reflect patients in the estimation sample six months before certification (or the control month 0 for the control sample). % > 0 indicates the share of patients with a strictly positive value. Other columns reflect the mean, standard deviation, and 10th, 50th, and 90th percentiles. Physician Characteristics are also averaged across patients.

Table A.2: Means by Patient Type

	Patients	Share	Age	Chronic	Spend	FFS Rate	Hours
1	147,775	0.115	10.484	0.000	2.646	32.973	0.081
2	96,503	0.075	32.094	0.099	5.080	47.685	0.107
3	83,275	0.065	40.384	0.122	5.765	45.822	0.126
4	54,410	0.042	37.941	0.055	8.752	45.807	0.192
5	65,015	0.051	41.193	0.001	9.331	46.662	0.201
6	51,919	0.040	43.938	0.041	10.248	46.466	0.222
7	50,825	0.039	59.143	0.501	11.671	47.772	0.246
8	35,968	0.028	66.521	0.760	15.302	45.837	0.336
9	33,473	0.026	59.451	1.000	18.823	48.723	0.388
10	24,200	0.019	72.333	1.000	25.271	50.351	0.504

Notes: Summary statistics reflect patients' monthly totals six months before certification in the estimation sample. Monetary measures are in USD. Hours are total reimbursement divided by a wage index.

Table A.3: Distribution of Patient Severity

	Estimate	Std. Err.
Patient Type 2	0.039	(0.001)
Patient Type 3	0.053	(0.001)
Patient Type 4	0.083	(0.001)
Patient Type 5	0.091	(0.001)
Patient Type 6	0.092	(0.001)
Patient Type 7	0.091	(0.001)
Patient Type 8	0.109	(0.001)
Patient Type 9	0.111	(0.001)
Patient Type 10	0.129	(0.002)
February	0.030	(0.001)
March	0.011	(0.001)
April	0.020	(0.001)
May	0.010	(0.001)
June	0.018	(0.001)
July	0.014	(0.001)
August	−0.059	(0.001)
September	0.013	(0.001)
October	0.017	(0.001)
November	0.017	(0.001)
December	0.018	(0.001)
$\log(1 + m_{t-1})$	0.024	(0.000)
$m_{t-1} = 0$	0.050	(0.001)
Cancer	0.010	(0.002)
Diabetes	0.028	(0.002)
COPD	0.031	(0.002)
Asthma	0.018	(0.002)
CVD	0.035	(0.002)
1+ Chronic Illness	0.014	(0.002)
2+ Chronic Illnesses	−0.005	(0.002)
Female	0.001	(0.000)
Disability Receipt	0.055	(0.001)
Income Percentile	−0.013	(0.001)
Recent Acute ER Visit	0.022	(0.001)
Recent Acute ER Visit 2+	0.032	(0.001)
Time Trend	0.009	(0.002)
New Patient	0.006	(0.001)
$\log \sigma_\lambda$	−0.389	(0.003)
$P(\lambda > 0) : d_0$	−3.389	(0.019)
$P(\lambda > 0) : d_1$	11.462	(0.132)

Notes: This table shows model estimates with asymptotic standard errors calculated using the approximate Hessian. Unobserved patient severity is distributed $\ln \lambda \sim N(\beta_\lambda X_\lambda, \sigma_\lambda) \mid \lambda > 0$ and $Pr(\lambda > 0) = f(d_0 + d_1 \beta_\lambda X_\lambda)$, where $f(z) = \frac{\exp z}{1 + \exp z}$. The first set of estimates corresponds to β_λ .

Table A.4: Test for Selection on Unobserved Physician Heterogeneity

	Certified	Non-Certified	Certified and Non-Certified		
	(1)	(2)	(3)	(4)	(5)
$E[m]$	1.041*** (0.002)	1.025*** (0.005)	1.032*** (0.002)	1.041*** (0.002)	1.087*** (0.003)
$E[m] \times \text{Control}$				-0.016*** (0.005)	-0.018*** (0.005)
Control				-0.001 (0.001)	
Female					-0.013*** (0.001)
Age					-0.000*** (0.000)
Chronic Illnesses					-0.021*** (0.001)
Intercept	-0.007*** (0.001)	-0.008*** (0.001)		-0.007*** (0.001)	
Physician FEs			✓		✓
Observations	2013672	385416	2399088	2399088	2399088
R ²	0.113	0.108	0.114	0.112	0.114

Notes: All regressions use observed treatment intensity as the dependent variable. The control (Non-Certified) sample is constructed identically to the main estimation (Certified) sample, except that the starting pool of physicians is a random subset of those that never become certified. The last three columns pool both samples. $E[m]$ is calculated based on parameter estimates given observable characteristics. Control is an indicator for the control sample.

Table A.5: Counterfactual Outcomes with Perturbations

	$\Delta SS_{Efficient}$	$\Delta SS_{Uniform}$		ΔSS_{Menu}		Menu $\succ$ Uniform
	Level	Level	Share of Eff.	Level	Share of Eff.	
Baseline	8.396	2.548	0.303	2.714	0.323	✓
$0 \times Var(c)$	7.885	2.122	0.269	2.464	0.313	✓
$\frac{1}{2} \times c$	3.423	2.183	0.638	2.184	0.638	✓
$2 \times c$	5.560	1.194	0.215	1.332	0.240	✓
$2 \times Var(c)$	15.123	2.361	0.156	2.361	0.156	
$0 \times Var(\alpha)$	8.664	2.606	0.301	2.921	0.337	✓
$\frac{1}{2} \times \alpha$	5.838	2.005	0.343	2.040	0.349	✓
$2 \times \alpha$	11.188	2.791	0.249	3.178	0.284	✓
$2 \times Var(\alpha)$	9.978	2.327	0.233	2.327	0.233	
$0 \times Var(\gamma)$	8.645	2.564	0.297	2.652	0.307	✓
$\frac{1}{2} \times \gamma$	2.892	0.881	0.305	0.933	0.322	✓
$2 \times \gamma$	22.371	5.519	0.247	6.030	0.270	✓
$2 \times Var(\gamma)$	8.733	2.542	0.291	2.733	0.313	✓
Uncorrelated c, α, γ	10.215	2.117	0.207	2.176	0.213	✓
Drop Outliers of c, α, γ	8.993	2.576	0.286	2.802	0.312	✓
$\frac{1}{2} \times Var(\theta_k), \theta_k \in c, \alpha, \gamma$	8.416	2.721	0.323	2.998	0.356	✓
$0 \times Var(\gamma), 0 \times Var(\alpha)$	8.680	2.763	0.318	2.991	0.345	✓
$0 \times Var(c), 0 \times Var(\alpha)$	7.622	2.466	0.324	2.819	0.370	✓
$0 \times Var(c), 0 \times Var(\gamma)$	8.318	2.124	0.255	2.421	0.291	✓
$\frac{1}{2} \times \sigma_\lambda$	6.446	1.732	0.269	1.803	0.280	✓
$2 \times \sigma_\lambda$	23.791	5.530	0.232	6.456	0.271	✓
$\frac{1}{2} \times \alpha_G$	4.449	1.324	0.298	1.310	0.294	
$2 \times \alpha_G$	16.599	4.991	0.301	5.667	0.341	✓
Add Control Sample	9.681	4.010	0.414	4.161	0.430	✓
Constrain Capacity	17.524	2.063	0.118	4.376	0.250	✓
Exclude Part-Time Physicians	8.781	2.559	0.291	2.730	0.311	✓
Only Urban Physicians	8.374	2.561	0.306	2.737	0.327	✓
Only Rural Physicians	9.360	2.644	0.282	2.788	0.298	✓
Alt. Health Parameterization	8.426	2.561	0.304	2.737	0.325	✓

Notes: This table compares key outcomes between counterfactual contract menus. All outcomes are based on ex-ante expectations over patient-months using estimated distributions of G and F , weighted across physicians by enrollment. Outcomes are summarized by the change in social surplus, defined as the change in health production versus pre-certification minus the change in expenditure versus post-certification. Share of Eff. divides the change in levels of social surplus for the optimal menu by the change in levels for efficient contracts. Relative to Table 4 (included as "Baseline"), each row perturbs one or more parameters before repeating counterfactual analyses. The parameters are marginal cost c , altruism α , productivity γ^{-1} , standard deviation of the log patient severity σ_λ , and altruism of the regulator α_R . $0 \times Var(c)$ fixes c at the sample mean. $\frac{1}{2} \times c$ multiplies c by 0.5 for all physicians. $2 \times Var(c)$ uses the following function: $f(c) = \bar{c} + \sqrt{2} \times (c - \bar{c})$. Outliers are below the 1st percentile or above the 99th of c , α , or γ . In one perturbation, I impose a capacity constraint on simulated hours per physician-month and approximate the shadow cost of capacity (see Appendix A.1 for details). Rural physicians are in municipalities with low centrality indexes. Part-Time physicians spend less than 25 hours per week with patients in the six months before certification.

Table A.6: Test for Patient Sorting

	Predicted Health (SDs) (1)	Switch (2)	Hospitalization (3)	Mortality (4)
Post-Certification	0.133*** (0.044)			
Predicted Health (SDs)		−0.001 (0.004)	−0.003*** (0.001)	−0.001* (0.001)
Age	−0.014*** (0.002)	−0.001*** (0.000)	−0.000*** (0.000)	−0.045*** (0.002)
Asthma	−0.200*** (0.075)	−0.005 (0.009)	0.052*** (0.014)	0.016 (0.015)
Age ²	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)
Cancer	−0.183*** (0.070)	0.006 (0.008)	−0.025* (0.014)	0.072*** (0.015)
COPD	−0.291*** (0.087)	0.033*** (0.009)	0.181*** (0.015)	0.108*** (0.016)
CVD	−0.341*** (0.089)	0.007 (0.008)	−0.034** (0.014)	0.025* (0.015)
Diabetes	−0.400*** (0.113)	−0.004 (0.008)	−0.036*** (0.014)	0.035** (0.015)
Female	−0.164*** (0.033)	0.009** (0.004)	−0.001 (0.001)	−0.001 (0.002)
Income Percentile	0.099*** (0.034)	0.022*** (0.005)	−0.010*** (0.001)	0.085*** (0.006)
1+ Chronic Illness	−0.111 (0.086)	−0.009 (0.008)	0.038*** (0.014)	−0.037** (0.015)
2+ Chronic Illnesses	−0.046 (0.069)	−0.000 (0.009)	0.040*** (0.014)	−0.032* (0.017)
Observations	2583264	215272	215272	54192
R ²	0.102	0.022	0.043	0.133
Outcome mean	−0.013	0.060	0.021	0.042

Notes: This table shows estimates of the correlation between patients' model-predicted health production and outcomes of interest measured after the estimation sample. Health production is normalized to standard-deviation units within the estimation sample. All specifications include year and calendar month fixed effects and cluster at the physician level. Column (1) includes the entire spell and regresses expected health production given parameter estimates on an indicator for months after specialization and patient demographics. Columns (2)-(4) are cross-sectional regressions using expected health production as the treatment variable of interest. The dependent variables are an indicator for switching to a new physician within 6 months, an indicator for an avoidable hospitalizations in the next 12 months, and an indicator for mortality within the next 24 months. Column (4) includes patients over 45 years old. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level.

Table A.7: Counterfactual Outcomes by Physician Location

Physicians		Efficient Contracts		Menu of Contracts		
Type	Share	$\Delta E[h(m)]$	$\Delta E[p m + b]$	$\Delta E[h(m)]$	$\Delta E[p m + b]$	$\Delta E[V(p)]$
Most Urban:	1	0.11	6.09	1.72	2.10	2.18
	2	0.31	8.90	2.30	3.08	2.42
	3	0.34	7.32	1.99	2.65	2.23
	4	0.16	9.22	2.46	2.57	2.15
	5	0.04	11.11	2.58	3.43	2.51
Most Rural:	6	0.04	13.50	2.69	4.36	2.84

Notes: This table shows average outcomes for efficient (personalized) contracts and the optimal menu of contracts, disaggregated across groups of physicians (rows). Physicians are grouped by the centrality index of their municipality. $\Delta E[h(m)]$ represents the change in health production relative to the status quo, for efficient contracts and the optimal menu of contracts. Likewise, $\Delta E[p m + b]$ represents incremental expected expenditure and ΔEV represents incremental expected indirect utility. Outcomes are averages across patients within each group, measured in USD. I assume that the less than 1 percent of physicians who do not have a linked municipality are in the most urban category.

A.1 Income Effects and Capacity Constraints

This section extends the main model to incorporate nonlinearity in the effect of the fee-for-service rate on treatment intensity via two mechanisms. With income effects, there is utility from leisure, or equivalently, disutility from total workload. With capacity constraints, total treatment intensity has an upper bound. Both mechanisms imply that counterfactually increasing the fee-for-service rate would have muted effects relative to the baseline model.

Income effects introduce complementarity between the treatment intensity decisions of various patients. Increasing the treatment intensity for one patient may increase the marginal utility of leisure, lowering treatment intensity for other patients. To tractably model complementarity, I assume that a physician will treat each patient slightly less intensively if that physician expects to work many hours over the next month treating other patients.⁶⁹ Physicians with a large number of patients (“enrollment”) or patients with high expected severity (“composition”) expect to work more hours. As before, for each patient $i \in 1, \dots, N$, the health shock is realized only when that patient arrives, so the private objective includes

⁶⁹This assumption implies that patients arrive sequentially and only short-term future treatment intensity affects the marginal utility of leisure. Conditional on expected workload, treatment intensity is independent across patients. Alternatively, I could relax the assumption that the marginal utility of net income equals 1 at the cost of complicating the expression for physicians’ private gain from switching contracts.

an expectation over the disutility of workload:

$$EV(x; \lambda_i, F, \theta) = \max_{m_i \geq 0} x(m_i) - c(m_i) + \sigma E \left[l \left(\sum_{i'=1}^N m_{i'}^* \right) \mid F(\lambda_{i'}) \right] + \alpha h(m_i, \lambda_i), \quad (5)$$

In estimation, I assume quadratic preferences, $l(x) = -\frac{(x)^2}{2}$, and I substitute observed average treatment intensity for expected treatment intensity since the two should coincide at true parameters with many patients. The privately optimal level of treatment intensity becomes:

$$m^*(p, \lambda, (N-1)\bar{m}) = \max\{0, \frac{p - c - \sigma(N-1)\bar{m} + \alpha\gamma\lambda}{\alpha + \sigma}\}. \quad (6)$$

The new parameter σ is identified by variation within-physician across-time in expected workload from changes in enrollment. Like the fee-for-service rate, changes in enrollment must be exogenous to physician type and unobserved patient severity. Observing each physician with the same incumbent patients for multiple periods at each fee-for-service rate helps with identification. Consistent with prior studies that find treatment intensity increases in the fee-for-service rate, likelihood ratio tests fail to find evidence of strong income effects.⁷⁰ Although average treatment intensity sometimes does not increase after certification, high altruism and dispersion in unobserved patient severity better explain this pattern than income effects.

If capacity constraints sometimes bind, then over a long period, with idiosyncratic variation in enrollment, composition, and realized severity, some physicians' monthly total treatment intensity should bunch near the maximum. I instead find that the distribution of treatment intensity relative to a physician-specific maximum is relatively smooth near the maximum over 120 months (See A.13). Moreover, the main counterfactual results are also robust to imposing capacity constraints (See Table A.5). To implement that robustness check, I reduce each treated patient's treatment intensity by a fixed amount per physician-month: excess total hours per treated patient, where excess total hours is the difference between unconstrained total hours and capacity. This approach follows from adding a capacity constraint, $\sum_i m_i^*(p) \leq \bar{M}$, to the objective which implies the first-order condition: $m^*(p) = \max\{0, \frac{p-c-\mu_c}{\alpha} + \gamma\lambda\}$, where μ_c is the shadow cost of capacity. Substituting this condition into the capacity constraint gives $\mu_c = \alpha \left(\frac{\sum_i \max\{0, \frac{p-c-\mu_c}{\alpha} + \gamma\lambda\} - \bar{M}}{N_{jt} Pr(m^*(p) > 0)} \right)$. The fixed point μ_c may not converge with quadrature, so I approximate it as $\hat{\mu}_c = \alpha \left(E[m_{ijt}^0 \mid m_{ijt}^0 > 0] - \frac{\bar{M}}{N_{jt} Pr(m_{ijt}^0 > 0)} \right)$ where $m_{ijt}^0 = \max\{0, \frac{p-c}{\alpha} + \gamma\lambda\}$ is the unconstrained treatment intensity.⁷¹

⁷⁰In estimation, I search over positive scaled values of σ .

⁷¹Two further adjustments help limit approximation error. First, I bound the denominator below by 1.

A.2 Counterfactual Extensions

Even when considering more flexible contract structures, a menu of linear contracts tends to dominate a uniform contract, and the gap between a menu and personalized contracts under full information remains large. First, instead of requiring all physicians to choose from a single menu of contracts (“Baseline”), I derive multiple menus of contracts that incorporate observed heterogeneity. Offering ten menus – one for each bin of patients with similar characteristics – does not lead to larger welfare gains.⁷² One explanation is that within a patient type, there is limited variation in average severity. Bundling the entire patient panel helps facilitate correlation across physicians between average severity and other traits (c, α) like in Section 2.2.

Second, Table A.5 shows that counterfactual outcomes are similar to Baseline when separating the analysis between urban and rural patients.⁷³ The limited benefit of regional contracts is surprising because in the status quo, regional variation is one of the few exceptions to nationally uniform reimbursement.⁷⁴ Baseline (national) contracts might perform relatively well because within-region physician heterogeneity features weaker favorable correlations than across-region heterogeneity. Consistent with prior literature, Table A.7 shows that health disparities among rural patients remain a pressing concern. Relative to the status quo, eliminating information asymmetry about physicians improves patient health by \$13 for the most rural patients and \$6 for the most urban patients. A national menu of contracts helps narrow the gap in health, but only by a fraction. Similarly, regional menus achieve less than one-third of first-best welfare gains.

Third, relaxing the linear structure of contracts does not increase the welfare achievable with a uniform contract. Although such contracts are rare in healthcare settings and perhaps difficult to implement, larger welfare gains may be possible when revenue is a flexible function of treatment intensity. For example, after a large amount of treatment, the marginal return to health may be small, and low marginal reimbursement would limit relatively inefficient spending. I find that the optimal nonlinear uniform contract substantially improves patient

Second, I use 0.5 as a threshold for m_{ijt}^0 in the denominator to avoid over-correcting based on mass near zero treatment intensity. This threshold represents approximately the 90th percentile of status quo treatment intensity.

⁷²I use the same procedure as before, except that each counterfactual fee-for-service rate is a level rather than a multiple of the status quo. Aggregating across menus, the average incremental social surplus is \$2.58 per patient-month versus \$2.74 for the baseline single menu.

⁷³I use the same procedure as Baseline, but separately for rural and urban physicians. Approximately one-fourth of physicians are classified as rural because they practice in a low-centrality municipality.

⁷⁴For example, in Norway, physicians in very small municipalities receive additional payments per registered patient. In the United States, Medicare reimbursement adjusts for rural status, the share of low-income patients, and a local wage index.

health relative to a menu of contracts. The gain is around half as large as from efficient linear contracts. However, the participation constraint requires large increases in expenditure on base payments, so the net gains to welfare are small: 23 percent of first-best. Without base payments, 56 percent of physicians are worse off, and these losses represent up to 5 percent of status quo revenue. Appendix A.3 provides details on deriving the nonlinear contract (extending Gaynor, Mehta and Richards-Shubik, 2023), and compares distributional outcomes across counterfactuals. For each segment along a grid of treatment intensity, the optimal marginal payment maximizes incremental health production net of incremental private costs, among patients with marginal treatment intensity. I find that the optimal nonlinear contract is approximately linear beyond low levels of treatment intensity (Figure A.15), redistributing away from patients with low severity to most other patients with relatively high severity (Figure ??).

Institutional differences may explain the different impacts of a non-linear uniform contract in this setting relative to Gaynor, Mehta and Richards-Shubik (2023). With primary care and the large estimated dispersion in unobserved patient severity, there does not seem to be a narrow range of medically appropriate treatment intensity for a non-linear contract to target. Moreover, my estimates imply that marginal health production is nearly universally positive, so decreasing treatment intensity is not generally efficient. In Gaynor, Mehta and Richards-Shubik (2023), more than half of observed treatment intensity was high enough to damage health because medication dosage exceeded a known cutoff.⁷⁵

In addition to contract flexibility, the regulator might further improve patient health through policies that complement contracts by shifting the allocation of patients across physicians. For example, in Table 5, the decomposition across physician and patient types suggests that perhaps high-severity patients should not be registered with high-cost low-altruism physicians. The combination of relatively high severity, high cost, and low altruism partly explains the larger impacts of counterfactual contracts in rural areas (see Table A.7). In reality, physicians decide where to establish a practice and most will move at least once during their career. At these times, contract heterogeneity could incentivize different location choices. For example, it might induce better match quality to increase the base payment for a high-fee-for-service contract in areas where nearby patients have relatively high observed severity. Alternatively, the regulator could incentivize patients to switch to under-subscribed high-quality physicians. While this question is beyond the scope of the current work, it may be a fruitful path for future research: combining efficient reimbursement rates with optimal patient switches can increase incremental social surplus by 17 percent relative to efficient

⁷⁵These characterizations mostly refer to Figure 3 in that paper, which is based on a patient with median observed severity.

reimbursement rates alone.⁷⁶

So far, while fixing the distribution of physicians, information asymmetry remains costly even after adding contract flexibility – perhaps the regulator could further improve patient health through complementary long-run investments that alter the distribution of physician heterogeneity. At reasonable reimbursement rates, the regulator prefers a physician with low cost of effort, high altruism, and high productivity. Public subsidies for support staff or telehealth might lower cost of effort; performance benchmarks might increase altruism, and promoting long-term patient-physician relationships might increase productivity via soft knowledge. For example, performance benchmarks can increase information and facilitate learning, particularly about past patients that have since left the list. Physicians currently do not observe all the long-term impacts of treatment, like utilization and avoidable hospitalizations.

A.3 Optimal Nonlinear Uniform Contract

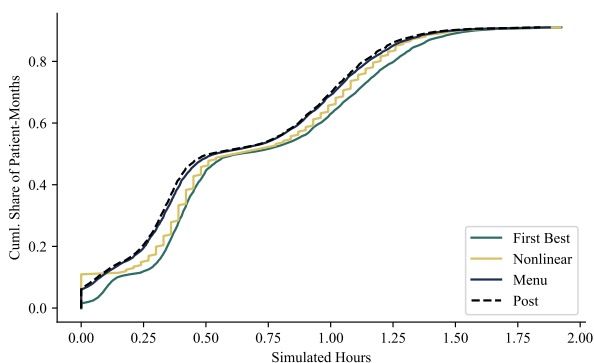
This paper primarily investigates contracts in which revenue is a linear function of treatment intensity. A linear structure nests the ways healthcare providers are typically reimbursed in most settings. Linear contracts may be common because they are relatively easy to implement and understand.⁷⁷ However, larger welfare gains may be possible when revenue is a flexible function of treatment intensity. For example, after a large amount of treatment, the marginal return to health may be small, so low marginal reimbursement can limit relatively inefficient spending.

The optimal nonlinear uniform contract performs about half as well as efficient linear contracts in terms of improving patient health, but the participation constraint requires large increases in expenditure, so the gains to social surplus are small. Figure A.15 shows that relative to the best linear uniform contract, the nonlinear contract lowers the marginal rates of low levels of treatment intensity and increases the marginal rates of high levels of treatment intensity. Figure A.14 compares the distribution of outcomes across patients and counterfactuals and the status quo. The nonlinear contract generally increases treatment intensity except for decreases among patients with very low (positive) severity. This dis-

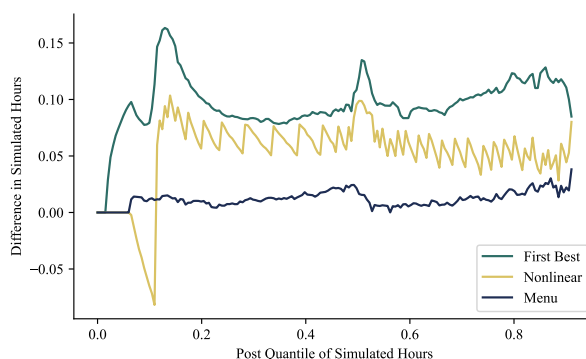
⁷⁶This exercise involves a stylized example of two vertically differentiated physicians at the 10th and 90th percentile of (initial) efficient fee-for-service rates. I begin by counterfactually assigning both physicians the average patient distribution, corresponding fee-for-service rates, and average enrollment. I alternate between searching for first-best contracts (fixing patient lists) and searching for a patient swap (fixing contracts), both optimizing average social surplus. This method maintains the initial number of patients per physician and converges after 53 percent of patients have switched.

⁷⁷For example, Norway uses a survey of physicians' costs to inform service-level uniform reimbursement rates. When nonlinear contracts exist, they tend to be simple like caps or outlier payments.

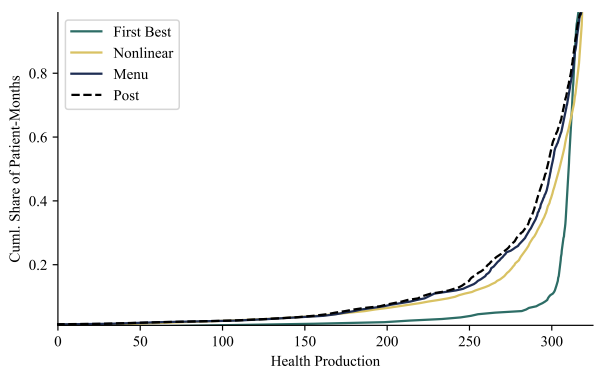
Figure A.14: Distribution of Treatment Intensity, Health Production, and Expenditure Across Contract Types



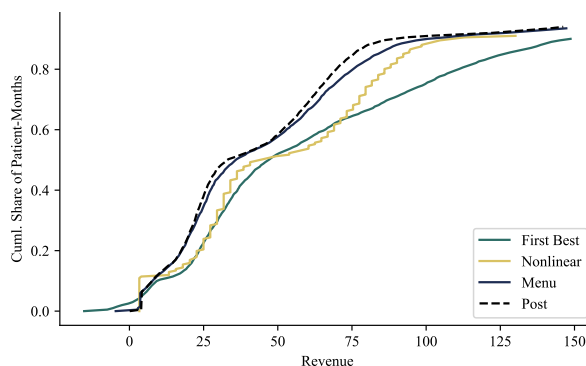
(a) Cumulative Distribution



(b) Difference in Cumulative Distributions



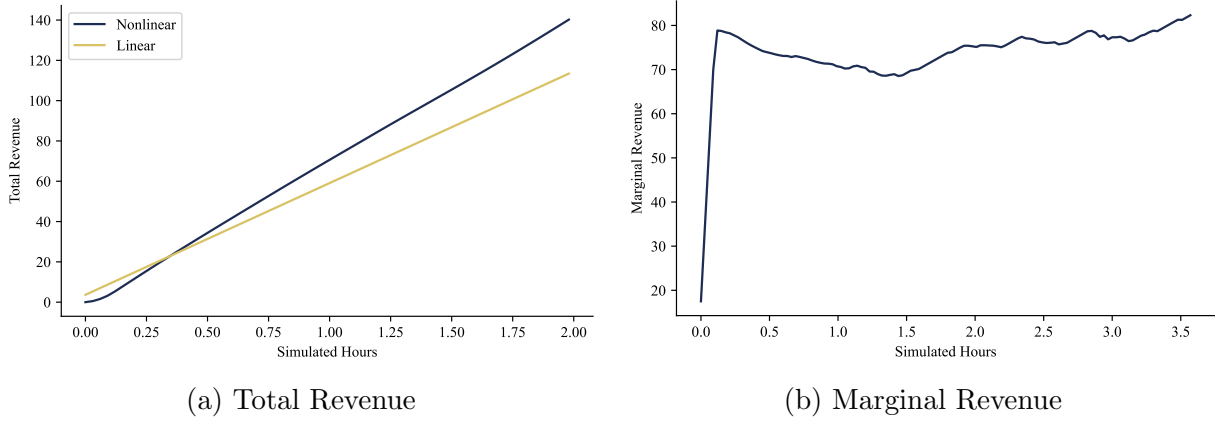
(c) Cumulative Distribution of Health Production



(d) Cumulative Distribution of Expenditure

Notes: Panels (a) and (b) compare the distribution of treatment intensity, and panels (c) and (d) compare the distribution of health production and expenditure, under the optimal nonlinear uniform contract, a menu of linear contracts, and first-best (full information) linear contracts. Panel (b) illustrates how treatment intensity compares to the status quo, under the higher reimbursement rate of certified physicians, by subtracting corresponding quantiles, e.g., the 5th percentile of treatment intensity under first-best contracts minus the 5th percentile of treatment intensity under the status quo. Post-Cert. indicates the status quo with a high reimbursement rate from certification. First Best indicates efficient linear contracts under full information. These plots condition on a positive health shock.

Figure A.15: Optimal Nonlinear Uniform Contract



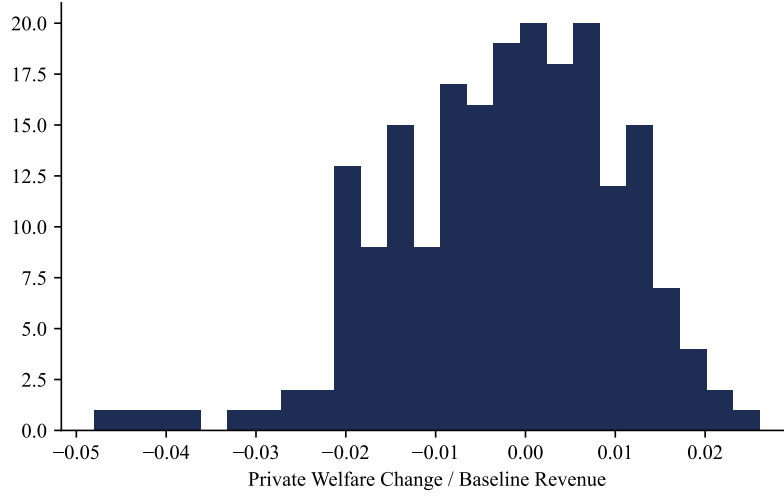
Notes: These plots illustrate the optimal nonlinear uniform contract which provides revenue as a flexible function of treatment intensity (along the x-axis). Linear indicates the optimal linear uniform contract including the base payment. In this figure, the nonlinear contract does not include the base payment.

tribution of treatment intensity more closely resembles efficient linear contracts with the nonlinear uniform contract than with menu of linear contracts. Fee-for-service expenditure is also generally lower. However, without a base payment, the nonlinear uniform contract is not directly comparable to other counterfactuals because the nonlinear contract redistributes from physicians towards patients. Figure A.16 illustrates this tradeoff: although physicians are on average equally well off under the nonlinear contract relative to the status quo, 56 percent are individually worse off, and these losses represent up to 5 percent of status quo revenue. Physicians with losses tend to have high productivity and low patient severity, i.e., less initial under-treatment. If these physicians eventually exit the system, the risk of under-treatment for unmatched patients grows. For all physicians to be weakly better off, as required in other counterfactuals, the nonlinear contract needs \$3.32 in base payments, which is slightly higher than average base payments under the menu. After this adjustment, the gain in social surplus is only 23 percent of first-best while the menu of linear contracts achieves 34 percent.⁷⁸

I use a demand profile approach to derive the nonlinear contract, similar to Gaynor, Mehta and Richards-Shubik (2023), while also drawing on intuition from Chade et al. (2022). The demand profile approach approximates the global design problem – avoiding the need to simultaneously for each level marginal reimbursement across the continuum of treatment

⁷⁸Alternatively, one could moderate the participation constraint by incorporating physician exit: if indirect utility is lower than the threshold, then the corresponding patients have zero treatment intensity and zero public expenditure. However, this form of the constraint would only negligibly change counterfactual outcomes. Across counterfactuals, health production net of expenditure exceeds health production with zero treatment intensity by more than the base payment for either all physicians or all but one.

Figure A.16: Some Physicians Worse Off with the Optimal Nonlinear Uniform Contract



Notes: This plot shows that many physicians experience moderate losses in private indirect utility under the optimal nonlinear uniform contract, as a share of status quo revenue. The reference point is each physician's pre-certification status quo. The y-axis is a count of physicians. The underlying data is an expectation over patients the distribution of severity.

intensity. For each interval of treatment intensity $[m_t, m_{t+1})$, I find the corresponding reimbursement rate p_t that maximizes incremental health production net of incremental private costs:⁷⁹

$$E_{\theta, \lambda} [(\alpha_R + \alpha)(h(m^*(p_t), \gamma\lambda) - h(m_t, \gamma\lambda)) - c(m^*(p_t) - m_t) \mid m^*(p_t) \geq m_t]$$

The interior of the expectation is a transformation of the social objective. Recall that the regulator maximizes expected health production subject to budget and participation constraints as well as privately optimal treatment intensity m^* which depends on the contract x , type θ , and patient severity. Based on equivalence of the first-order conditions after fixing shadow costs, I maximize a weighted sum of health production, private indirect utility, and public expenditure.⁸⁰ In the objective's conditional expectation, I focus on patients and physicians who contribute to incremental social surplus, i.e., those whose treatment intensity varies with p_t . I also restrict the space of contracts so that marginal reimbursement $p(m)$ only crosses effective marginal cost $c - \alpha h_m(m, \gamma\lambda)$ for a unique value of treatment intensity.⁸¹

⁷⁹Each interval spans approximately 2 simulated minutes of treatment intensity between 0 and 2 simulated hours.

⁸⁰I fix the shadow costs of expenditure (μ_B) and participation ($\mu_{P, \theta}$ at $\frac{1}{\alpha_R}$ before rescaling the objective by α_R . Equivalently, α_R is the regulator's altruism and the regulator is willing to sacrifice \$1 of expenditure to either increase health production or private indirect utility by \$1, so public expenditure and private revenue add to zero.

⁸¹Equivalently, the physician objective is concave and single-peaked for all physicians and patients. Marginal

B Data and Estimation Details

B.1 Data Sources

I use several data sources to construct the estimation sample. The Norwegian Control and Payment of Health Reimbursements Database (KUHR) tracks reimbursement for outpatient claims organized at the level of bill line, i.e., reimbursement code, and identifies most patients and physicians. The Norwegian Patient Registry (NPR) is a database of reimbursement for inpatient claims organized at the level of encounter. I use ICD-10 and ICPC-2 codes from both sources to classify chronic illness. I identify avoidable hospitalizations following Table A1 from Page et al. (2007). Base payments come from a basic subsidy rate dataset. Various datasets from the Norwegian GP Registry identify periods when patients are registered to patient lists and when physician are contracted to provide care to those patient lists. The physician-list dataset also identifies contract details: the maximum number of registered patients and indicators for shared lists and fixed-salary reimbursement. The data providers created specific instances of identifiers for physicians, lists, and patients to link datasets and convert periods into monthly panels. Physicians' birth date, gender, and birth country come from a personnel file. Patients' birth date, gender, disability payment receipt, and income come from tax records.

B.2 Construction of Treatment Intensity

I classify each patient into an observed type based on the combination of gender, 5-year age bins, and indicators for first and second prior chronic diagnosis, including cancer, diabetes, COPD, CVD, or asthma. I sort these 108 initial groups based on average reimbursement and further aggregate them into 10 types. Each aggregated type represents approximately 10 percent of aggregate spending in the estimation sample because treatment intensity is distributed approximately log-normally. The lowest type includes 23 percent of patient-months and the highest type represents 4 percent of patient-months.

For each patient type, I use all Norwegian patients to calculate the average bundle of services received and the average hours required to provide that bundle. I attribute time to encounters and reimbursement codes based on the share of reimbursement within an hour in the utilization data, e.g., 1-2 pm on January 1, 2010. I multiply each non-certification reimbursement code by the current administrative reimbursement rate. I average across

revenue cannot increase at a greater rate than marginal health production: $p_t - p_{t-1} < \alpha \forall t, \alpha$. I also focus on contracts where revenue is weakly increasing in treatment intensity. Neither of these constraints binds at the solution.

codes, weighting by the number of lines per patient type per month. After certification, this numerator also includes current certification supplementary payments for an average number of visits per patient type. Finally, I divide by average hours per patient-type to calculate the simulated wage p_{kt} , i.e., the reimbursement per hour a physician would receive for providing the average bundle of services to a patient of type k in month t . Treatment intensity m_{ijt} equals patient-month fee-for-service revenue divided by the simulated wage and roughly corresponds to hours of treatment per patient-month (“simulated hours”).

B.3 Counterfactual Analysis

This section reviews the technical assumptions underlying counterfactual analysis. I first describe the process for quantifying counterfactual outcomes given contracts. Then, I detail the algorithms that identify each set of contracts: efficient contracts, the optimal uniform contract, the optimal two-contract menu, and the optimal menu of contracts.

I measure all counterfactual outcomes as ex-ante expectations over registered patients of certified PCPs. I simulate patient severity for 60 patient simulants for each physician in the sample: 10 patient types multiplied by 6 quadrature nodes. For each of the 10 patient types per physician, I use averages of β_λ and $Pr(\lambda > 0)$, which aggregate over in-sample patients’ observed characteristics like chronic illnesses and age. From the physician’s first-order condition, treatment intensity is a function of simulated severity, estimated physician type, and contract. Likewise, indirect utility is a function of physician type, predicted treatment intensity, simulated severity, and the contract. Within a given menu, each physician’s privately optimal contract maximizes average indirect utility. Ex-ante expectations reflect three levels of aggregation.⁸² First, I average across quadrature nodes using quadrature weights to approximate the integral of normally distributed log patient severity. Second, I average across patient types, weighting by the observed number of patients in the estimation sample per physician. Third, I average across physicians, weighting by total registered patients six months before certification.

Health production per simulated patient equals $H - \frac{1}{2}\alpha_R(m^* - \gamma\lambda)^2$. α_R can be thought of as the regulator’s altruism or the inverse of the shadow cost of expenditure. I calibrate it with a revealed preference assumption. When setting supplementary reimbursement for certification, the regulator values incremental health production exactly as much as incremental expenditure. Expenditure equals $pm^*(p; \lambda, \theta) + b$, i.e., privately optimal treatment intensity multiplied by fee-for-service rates plus the base payment. I generally report incremental

⁸²When calculating expected indirect utility per physician per contract, I only aggregate over quadrature nodes and patient types.

expected health production which subtracts the pre-certification expected value.

To focus on the role of reimbursement in treatment intensity, I fix total registered patients, the share of patient types for each physician, pre-certification fee-for-service rates, and status quo base payments at values six months before certification. For example, this removes variation in patient severity from seasonality and the time trend, so counterfactual treatment intensity at post-certification fee-for-service rates will typically be higher than observed in the data. To be consistent, I simulate all post-certification outcomes following the same process as counterfactuals, using the immediate change in the fee-for-service rate.

I enforce budget and participant constraints in counterfactuals when possible. I assume post-certification expected expenditure is the budget. Likewise, for participation constraints, I use expected indirect utility during the sample period to construct physician-specific participation thresholds. Physicians continue to work throughout the sample period at those levels of indirect utility, so they might reasonably be expected to continue in counterfactuals. I aggregate participation constraints by requiring that the same share of physicians weakly prefer counterfactual contracts over the lesser of their pre- or post-certification contract.

I solve the regulator’s objective numerically for the set of physicians in my sample. All counterfactuals use a grid of 200 equally spaced points between 0.5 and 3.5. Each point reflects a multiple of pre-certification fee-for-service rates, which vary across physicians and patient types. The optimal uniform contract maximizes overall expected health production while satisfying global constraints. In other counterfactuals, the global budget constraint creates complementarity across physicians, and constrained maximization algorithms do not work well in this context. Instead, I enforce the participation constraints directly and search for contracts that maximize social surplus, i.e., incremental expected health production minus incremental expected expenditure.

I identify efficient contracts by solving physician-specific problems. I select the fee-for-service rate that maximizes a physician’s social surplus, then set the base payment so that the participation constraint binds. This solution is approximate because physicians have different numbers of patients and the weighted average of differences does not equal the difference of weighted averages. In some robustness checks, I take an additional step to enforce the global budget constraint. I lower the fee-for-service rate multiple by one grid point for one physician at a time to produce the smallest reduction in social surplus while lowering expenditure until the budget is slack.

I search for the optimal menu using a coordinate-wise price adjustment. I adjust one contract’s base payment at a time to maximize equilibrium incremental social surplus, holding other base payments fixed. Base payment updating repeats over the sequence of contracts until base payments converge. I initially restrict this sequence to contracts within 10 percent

of the optimal uniform contract's rate, expanding to an increasingly large neighborhood of the optimal uniform contract only as long as further search continues to improve social surplus, to avoid evaluating implausible contracts that no physician would select. For stability, I search over two grids of base payments rather than use an optimization routine. The first iteration uses a broad grid that spans all possible participation rates for the finite set of physicians. Each grid point is one physician's willingness to pay for a focal rate relative to the best uniform contract. The second iteration searches locally for improvements using a grid of quadrature nodes to avoid inframarginal base payments that do not affect participation. I enforce the participation constraint by always including the best uniform contract in the menu, but the global budget constraint is difficult to strictly enforce with this method, so I maximize health production net of expenditure and penalize increased expenditure over the budget. In particular, the objective is $\Delta E[h(m^*)|b(p)] - \min\{0, \Delta R\} + \max\{0, \Delta R\}^2$ where $R \equiv E[pm^* + b(p)|b(p)]$ and Δ subtracts the reference values from counterfactual outcomes.

C Derivations

C.1 Conditions for Efficient Self-Selection

Throughout this section, I continue to assume that local health returns to treatment intensity h' are (1) decreasing in treatment at a fixed rate ($\eta \equiv -h_{mm} > 0$); (2) increasing in effective severity $\gamma\lambda$ at a fixed rate ($\beta \equiv h_{m,\gamma\lambda} > 0$); and (3) positive ($h_m > 0$) which via the physician first-order condition implies $p < c$ for the range of fee-for-service rates considered.

The regulator objective can be separated across physicians and written as:

$$\begin{aligned}
W_\theta(p) &\equiv E_\lambda [h(m_\theta^*(p), \gamma\lambda) - \mu_B (pm_\theta^*(p) + b)] \\
\partial_p W_\theta(p) &\equiv E_\lambda [h_m(m_\theta^*(p), \gamma\lambda) \partial_p m_\theta^*(p) - \mu_B (m_\theta^*(p) + p \partial_p m_\theta^*(p))] \\
&= E_\lambda [\partial_p m_\theta^*(p) \left(\frac{c-p}{\alpha} - \mu_B p\right) - \mu_B m_\theta^*(p)] \\
&= \underbrace{\partial_p m_\theta^*(p) \left(\frac{1+\alpha\mu_B}{\alpha}\right) (p_\theta^{FB} - p)}_{\equiv a_\theta(p)} - \underbrace{\mu_B E_\lambda[m_\theta^*(p)]}_{r_\theta(p)}
\end{aligned}$$

Rate p is not necessarily chosen by physician θ . I omit the participation constraint because I use this expression relative to a reference contract p_0 where b_0 is high enough for full participation. The first equality substitutes using the physician's first-order condition, and the second isolates the efficient rate while expanding the expectation by conditioning on physician type, using the fixed concavity of h so that $\partial_p m_\theta^*(p)$ does not depend on severity.

Physician θ has positive social returns to a higher rate when inframarginal rents $r_\theta(p)$

are outweighed by the allocative effect $a_\theta(p)$, which consists of the gap between the socially efficient for θ and the current rate, scaled by responsiveness and the relative (private vs. public) weights on health production. The decomposition reinforces the intuition that the social benefit of allocating a physician to a marginal higher rate is larger when that physician is relatively far below their efficient rate. With the optimal uniform contract, marginal social surplus is zero in expectation across physicians, i.e., $E_{\theta,\lambda}[\partial_p W_\theta(p)] = 0$, which means that $E_{\theta,\lambda}[a_\theta(p)] = E_{\theta,\lambda}[r_\theta(p)] = \mu_B E_{\theta,\lambda}[m_\theta^*(p)] > 0$.⁸³ Therefore some physicians have positive allocative effects, i.e., higher efficient rates, which raises the question of whether a menu could feasibly allocate these physicians to higher rates.

Suppose the regulator started with the optimal uniform contract (p_0, b_0) , where participation constraints are all met, and then gave all physicians the option to choose a marginally higher-rate contract $p_1 = p_0 + \epsilon > p_0$. From the envelope condition, physicians with relatively high expected treatment intensity will choose the new contract (p_1, b_1) : $\partial_p E_\lambda[V(p, b)] = E_\lambda[m_\theta^*(p)] > b_0 - b_1 \equiv b_\epsilon$. Putting these together, a menu increases social welfare if $\exists b_\epsilon$ such that

$$\begin{aligned} & E_\theta [a_\theta(p) - r_\theta(p) \mid E_\lambda[m_\theta^*(p)] > b_\epsilon] \\ &= E_\theta \left[\left(\frac{1+\alpha\mu_B}{\alpha^2|h_{mm}|} \right) (p_\theta^{FB} - p) - \mu_B E_\lambda[m_\theta^*(p)] \mid E_\lambda[m_\theta^*(p)] > b_\epsilon \right] > 0. \end{aligned} \quad (7)$$

This perturbation menu is efficient only if the high-treatment physicians who switch contracts also have high efficient rates ($p_\theta^{FB} > p$), so that the allocative effect outweighs the inframarginal rents. Rents are largest among switching physicians who have high baseline treatment intensity. The gap between efficient and uniform rates can be smaller if the scaling factor $\partial_p m_\theta^*(p) \left(\frac{1+\alpha\mu_B}{\alpha} \right)$ is large, e.g., due to generally low altruism.

A slightly stronger sufficient condition is that the net social and private gains from a higher rate are correlated across physician types:⁸⁴

$$\begin{aligned} \rho_\theta \equiv \text{Cov}_\theta(a_\theta(p) - r_\theta(p), E_\lambda[m_\theta^*(p)]) &> 0 \\ \text{Cov}_\theta(a_\theta(p), \underbrace{E_\lambda[m_\theta^*(p)]}_{\equiv s_\theta(p)}) &> \mu_B \text{Var}(E_\lambda[m_\theta^*(p)]) \end{aligned}$$

For each term, I use a first-order approximation with respect to $\theta = \{c, \alpha, \gamma\}$, around the

⁸³I assume a binding budget constraint and nondegenerate distribution of physician heterogeneity.

⁸⁴This condition is stronger because it applies to the entire distribution rather than only switchers. It is sufficient because social gains cannot be negative for all physicians at the optimal uniform contract. Moreover, social gains could be redefined to include reduced base payments on switchers ($\mu_B b_\epsilon > 0$), weakening the initial sufficient condition:

$E_\theta[a_\theta(p) - r_\theta(p) \mid s_\theta > b_\epsilon] > -\mu_B b_\epsilon$.

average physician $\bar{\theta}$. Let $a_k \equiv \partial_{\theta_k} a_\theta(p)$ and $s_k \equiv \partial_{\theta_k} E_\lambda[m_\theta^*(p)]$ for $\theta_k \in \{c, \alpha, \gamma\}$, i.e. “loadings,” then:

$$\begin{aligned} a_\theta(p) &\approx a_\theta(p; \theta = \bar{\theta}_k) + \sum_k a_k (\theta_k - \bar{\theta}) \\ s_\theta(p) &\approx s_\theta(p; \theta = \bar{\theta}_k) + \sum_k s_k (\theta_k - \bar{\theta}) \\ \text{Cov}_\theta(a_\theta - \mu_B s_\theta, s_\theta) &\approx \sum_j \sum_k (a_j - \mu_B s_j) s_k \text{Cov}(\theta_j, \theta_k). \end{aligned} \quad (8)$$

Note that $\text{Cov}(\theta_k, \theta_k) = \text{Var}(\theta_k)$, the expressions simplify because $a_\gamma = 0$, and, for constants $\bar{\theta}_j, \bar{\theta}_k$, $\text{Cov}(\bar{\theta}_j, \bar{\theta}_k) = 0$ and $\text{Cov}(\theta_j - \bar{\theta}_j, \theta_k - \bar{\theta}_k) = \text{Cov}(\theta_j, \theta_k)$.

To sign how covariance and variance of primitives enter this covariance between private and social gains, I calculate the partial derivatives of treatment intensity, then efficient rates, then the scaling factor in the allocative effect, and then aggregate terms. First, I total-differentiate the private first-order condition of $h_m(m_\theta^*, \gamma\lambda) = \frac{c-p}{\alpha}$ and using the interior-solution case of sufficiently large patient severity:

$$\begin{aligned} h_{mm} \partial_p m^* &= -\frac{1}{\alpha} &\Rightarrow \partial_p m^* &= \frac{1}{\alpha\eta} &> 0, \\ h_{mm} \partial_c m^* &= \frac{1}{\alpha} &\Rightarrow \partial_c m^* &= -\frac{1}{\alpha\eta} &< 0, \\ h_{mm} \partial_\alpha m^* &= -\frac{c-p}{\alpha^2} &\Rightarrow \partial_\alpha m^* &= \frac{c-p}{\alpha^2\eta} &> 0, \\ h_{mm} \partial_\gamma m^* + h_{m,\gamma\lambda} \lambda &= 0 &\Rightarrow \partial_\gamma m^* &= \frac{\lambda\beta}{\eta} &> 0. \end{aligned}$$

The corner solution for treatment intensity is 0, so expected treatment intensity, $s_\theta(p) \equiv E_\lambda[m_\theta^*(p)]$ has the same signs as above:

$$s_p > 0, \quad s_c < 0, \quad s_\alpha > 0, \quad s_\gamma > 0,$$

Next, I take partial derivatives of the efficient rate, relabeled as “target” $\tau_\theta \equiv c/(1 + \alpha\mu_B)$:

$$\tau_c = \frac{1}{1+\alpha\mu_B} > 0, \quad \tau_\alpha = \frac{-c\mu_B}{(1+\alpha\mu_B)^2} < 0, \quad \tau_\gamma = 0$$

I sign the scaling factor, $\kappa_\theta \equiv \partial_p m_\theta^*(p) \left(\frac{1+\alpha\mu_B}{\alpha} \right) = \frac{1}{\alpha^2\eta} + \frac{\mu_B}{\alpha\eta}$.

$$\kappa_\alpha = \frac{1}{\eta} \left(-2\alpha^{-3} - \alpha^{-2}\mu_B \right) = -\frac{2+\alpha\mu_B}{\eta\alpha^3} < 0, \quad \kappa_c = \kappa_\gamma = 0$$

Combining these terms within the allocative effect $a_k = \kappa_k(\tau_\theta - p) + \kappa\tau_k$:

$$a_c = \underbrace{\kappa_\theta}_{>0} \tau_c > 0, \quad a_\alpha = \underbrace{\kappa_\alpha}_{<0} (\tau_\theta - p) + \underbrace{\kappa_\theta}_{>0} \underbrace{\tau_\alpha}_{<0} < 0, \quad a_\gamma = 0$$

Note that $a_\alpha < 0$ is an additional assumption, albeit a weak one.⁸⁵ Next, I sign the loadings in $\partial_p W_\theta$, grouping terms from the allocative effect and inframarginal rents:

$$\underbrace{a_c}_{>0} - \mu_B \underbrace{s_c}_{<0} > 0, \quad \underbrace{a_\alpha}_{<0} - \mu_B \underbrace{s_\alpha}_{>0} < 0, \quad \underbrace{a_\gamma}_{=0} - \mu_B \underbrace{s_\gamma}_{>0} < 0$$

Then, multiplying each term by the respective s_k :

$$\begin{aligned} \partial_{\text{Var}(c)} \rho_\theta &= (a_c - \mu_B s_c) s_c < 0, & \partial_{\text{Var}(\alpha)} \rho_\theta &= (a_\alpha - \mu_B s_\alpha) s_\alpha < 0, \\ \partial_{\text{Var}(\gamma)} \rho_\theta &= -\mu_B s_\gamma s_\gamma < 0, & \partial_{\text{Cov}(c,\alpha)} \rho_\theta &= (a_c - \mu_B s_c) s_\alpha + (a_\alpha - \mu_B s_\alpha) s_c > 0, \\ \partial_{\text{Cov}(c,\gamma)} \rho_\theta &= (a_c - 2\mu_B s_c) s_\gamma > 0, & \partial_{\text{Cov}(\alpha,\gamma)} \rho_\theta &= (a_\alpha - 2\mu_B s_\alpha) s_\gamma < 0. \end{aligned}$$

Although physician heterogeneity is necessary for a menu to be efficient, i.e., dispersion in efficient rates, incremental dispersion weakens the alignment between private and social incentives. Likewise, alignment requires positive correlation between cost of effort and altruism, negative correlation between cost of effort and productivity (γ^{-1}), or positive correlation between altruism and productivity.

SIGNING a_α . $a_\alpha > 0$ implies that the best uniform rate is larger than the average physician's efficient rate by a wide margin. Loadings are evaluated at the average physician $\bar{\theta}$, so the relevant benchmark is $\bar{\tau}_\theta \equiv \frac{\bar{c}}{1+\bar{\alpha}\mu_B}$.⁸⁶

$$a_\alpha \equiv \partial_\alpha a_\theta = \frac{-2(c-p)}{\alpha^3 \eta} - \frac{-\mu_B p}{\alpha^2 \eta} = \frac{(2+\alpha\mu_B)p-2c}{\alpha^3 \eta} > 0 \iff p > \frac{2c}{2+\alpha\mu_B}$$

C.2 Likelihood

The likelihood is based on the random component of patient severity. Treatment intensity m may equal zero either because the underlying severity is zero or because it is too low for a privately optimal choice of $m > 0$. Since $\frac{\partial m}{\partial \lambda} > 0$, I can split cases based on $\tilde{\lambda}$, the minimum

⁸⁵I show below that $a_\alpha > 0$ only if $p > \frac{2\bar{c}}{2+\bar{\alpha}\mu_B} > \bar{\tau}_\theta \equiv \frac{\bar{c}}{1+\bar{\alpha}\mu_B}$. The best uniform rate would need to exceed the average physician's efficient rate by a wide margin. In the empirical application, the best uniform rate is well below the average efficient rate, so this case does not arise.

⁸⁶Recall that $\eta > 0$ is fixed and $\alpha > 0$, so $\alpha^3 \eta > 0$.

λ such that $m \geq 0$.

$$\begin{aligned} l(m \mid \theta, x, X_\lambda) &= l(m \mid \lambda \leq \tilde{\lambda})Pr(\lambda \leq \tilde{\lambda}) + l(m \mid \lambda > \tilde{\lambda})Pr(\lambda > \tilde{\lambda}) \\ &= 1[m = 0]Pr(\lambda \leq \tilde{\lambda}) + 1[m > 0]Pr(\lambda > \tilde{\lambda})Pr(\lambda = \lambda^{-1}(m) \mid \lambda > \tilde{\lambda}) \left| \frac{\partial \epsilon}{\partial \lambda} \frac{\partial \lambda}{\partial m} \right|. \end{aligned}$$

For $\tilde{\lambda} > 0$,⁸⁷ denoting the CDF of $\lambda \mid \lambda > 0$ as F_λ , the two-stage process for λ implies

$$Pr(\lambda > \tilde{\lambda}) = (1 - F_\lambda(\tilde{\lambda}))Pr(\lambda > 0) = 1 - Pr(\lambda \leq \tilde{\lambda}).$$

Under parametric assumptions,

$$\begin{aligned} \lambda^{-1}(m) &= \frac{m - \beta_0}{\beta_1} && \text{if } m > 0 \\ 0 \leq \lambda^{-1}(m) \leq \tilde{\lambda} &\equiv \max \left\{ 0, \frac{-\beta_0}{\beta_1} \right\} && \text{if } m = 0 \\ Pr(\lambda = \lambda^{-1}(m) \mid \lambda > \tilde{\lambda}) &= (1 - F_\lambda(\tilde{\lambda}))^{-1} \phi \left(\frac{\log \lambda^{-1}(m) - \beta_\lambda X_\lambda}{\sigma_\lambda} \right) \end{aligned}$$

where Φ and ϕ are the CDF and PDF of a standard normal and

$$\beta_0 \equiv \frac{p - c - \sigma(N-1)E[m']}{\alpha + \sigma}, \quad \beta_1 \equiv \frac{\alpha\gamma}{\alpha + \sigma} = \frac{\partial m}{\partial \lambda}, \quad \frac{\partial \epsilon}{\partial \lambda} = \frac{1}{\sigma_\lambda \lambda}.$$

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⁸⁷Recall that $Pr(\lambda > 0) = \frac{\exp(d_0 + d_1 \mu_\lambda)}{1 + \exp(d_0 + d_1 \mu_\lambda)}$ and $F_\lambda(\lambda) = \Phi \left(\frac{\log \lambda - \mu_\lambda}{\sigma_\lambda} \right)$ for $\mu_\lambda \equiv \beta_\lambda X_\lambda$.